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MITIGATING CATASTROPHIC FORGETTING IN OBJECT CLASSIFICATION VIA GAUSSIAN CLASS REPRESENTATIONS

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MITIGATING CATASTROPHIC FORGETTING IN OBJECT CLASSIFICATION VIA GAUSSIAN CLASS REPRESENTATIONS

Esta tesis propone un marco de Aprendizaje Incremental por Clases (CIL) libre de gradientes para mitigar el olvido catastrófico en el reconocimiento visual continuo, sin requerir retropropagación, optimización de parámetros ni almacenamiento de ejemplares. El sistema desacopla el aprendizaje de representaciones del modelado estadístico. El método acopla un modelo fundacional visual congelado y preentrenado mediante aprendizaje auto-supervisado (DINOv2) para extraer vectores de características estables de las imágenes, con una capa de clasificación generativa compuesta por modelos gaussianos multivariados locales e independientes por clase. Debido a que los parámetros de las clases históricas se congelan explícitamente, se elimina la deriva de la representación, logrando un olvido catastrófico cero por construcción.

Evaluado en Split CIFAR-100, Split ImageNet-R, Split Caltech-256 y en un benchmark conjunto de 30 tareas de largo horizonte, este enfoque determinista alcanza precisiones competitivas o superiores a las arquitecturas del estado del arte basadas en sugerencias (prompts) y de alta carga de optimización, al tiempo que añade un costo computacional secundario insignificante (¡0.3 GFLOPs). Además, una caracterización geométrica exhaustiva del espacio de incrustación (embedding) base valida formalmente los supuestos de normalidad gaussiana subyacentes. El análisis revela que la elección de la parametrización de covarianza óptima está gobernada por la densidad muestral y la dimensionalidad de las características: los estimadores diagonales robustos sobresalen bajo escasez de datos, mientras que una novedosa parametrización híbrida ofrece el mejor equilibrio entre precisión y calibración a medida que el presupuesto de muestras se expande.

MITIGATING CATASTROPHIC FORGETTING IN OBJECT CLASSIFICATION VIA GAUSSIAN CLASS REPRESENTATIONS

This thesis proposes a gradient-free Class-Incremental Learning (CIL) framework to mitigate catastrophic forgetting in continual visual recognition without requiring backpropagation, parameter optimization, or exemplar storage. The system decouples representation learning from statistical modeling. It pairs a frozen, pre-trained self-supervised visual foundation model (DINOv2) to extract stable image feature vectors with a generative classification layer composed of independent, per-class multivariate Gaussian models. Because historical class parameters are explicitly frozen, representation drift is eliminated, achieving zero catastrophic forgetting by construction.

Evaluated across Split CIFAR-100, Split ImageNet-R, Split Caltech-256, and a long-horizon 30-task Joint benchmark, this deterministic approach achieves accuracies competitive with or outperforming complex, prompt-based state-of-the-art architectures while adding negligible computational overhead (¡0.3 GFLOPs). Furthermore, a comprehensive geometric characterization of the foundation embedding space formally validates the underlying Gaussian assumptions. The analysis reveals that optimal covariance parameterization is governed by sample density and feature dimensionality: robust diagonal estimators excel under data scarcity, whereas a novel hybrid parameterization offers superior accuracy-calibration balance as sample sizes expand.

Abstract

Los modelos tradicionales de aprendizaje profundo colapsan ante flujos de datos no estacionarios, donde la incorporación de categorías novedosas exige un reentrenamiento computacionalmente exhaustivo o provoca el olvido catastrófico. Esta tesis presenta un marco de Aprendizaje Incremental por Clases (CIL) libre de gradientes que replantea el reconocimiento visual continuo como un problema de estimación de densidad en forma cerrada, en lugar de una optimización dinámica de parámetros. Al acoplar un codificador base congelado DINOv2 con modelos gaussianos multivariados locales por clase, estimados recursivamente a través de sus estadísticas suficientes, las nuevas categorías se integran de manera instantánea sin necesidad de retropropagación, almacenamiento de ejemplares o parámetros de optimización adicionales. Debido a que los parámetros de las clases históricas se congelan explícitamente, el marco elimina la deriva de la representación y logra un olvido catastrófico cero por construcción. Evaluado en Split CIFAR-100, Split ImageNet-R, Split Caltech-256 y en un benchmark conjunto de 30 tareas altamente heterogéneo, este enfoque determinista alcanza precisiones de clasificación competitivas o superiores a las arquitecturas del estado del arte basadas en sugerencias (prompts) y de alta carga de optimización, al tiempo que añade un costo computacional secundario insignificante (< 0.3 GFLOPs). Además, una caracterización estadística exhaustiva del espacio de incrustación (embedding) base —utilizando pruebas formales de normalidad univariada y multivariada, análisis espectral global y diagnósticos de distancia interclase de Bhattacharyya— valida empíricamente los supuestos de modelado fundamentales. Este análisis geométrico revela que la elección de la parametrización de covarianza óptima está estrictamente gobernada por la interacción entre la densidad muestral local y la dimensionalidad del espacio de características. Mientras que los estimadores diagonales robustos sobresalen bajo una severa escasez de datos al mitigar la deficiencia de rango, una parametrización híbrida que desacopla los términos de forma y volumen de la función de log-verosimilitud proporciona el mejor equilibrio entre precisión y calibración a medida que el presupuesto de muestras por clase se expande.

Abstract

Traditional deep learning architectures collapse under non-stationary data streams, where incorporating novel categories demands computationally exhaustive retraining or triggers catastrophic forgetting. This thesis presents a gradient-free Class-Incremental Learning (CIL) framework that reframe lifelong visual recognition as a problem of closed-form density estimation rather than dynamic parameter optimization. By pairing a frozen DINOv2 foundation encoder with localized, per-class multivariate Gaussian models estimated recursively via their sufficient statistics, new categories are integrated instantaneously without backpropagation, exemplar storage, or optimization parameters. Because historical class parameters are explicitly frozen, the framework eliminates representation drift and achieves zero catastrophic forgetting by construction. Evaluated across Split CIFAR-100, Split ImageNet-R, Split Caltech-256, and a highly heterogeneous 30-task Joint benchmark, this deterministic approach achieves classification accuracies competitive with or outperforming optimization-heavy, prompt-based state-of-the-art architectures, while adding negligible downstream computational overhead (< 0.3 GFLOPs). Furthermore, a comprehensive statistical characterization of the foundation embedding space—utilizing formal univariate and multivariate normality tests, global spectral analysis, and inter-class Bhattacharyya distance diagnostics—empirically validates the core modeling assumptions. This geometric analysis reveals that the choice of optimal covariance parameterization is strictly governed by the interplay between localized sample density and the ambient feature dimensionality. While robust diagonal estimators excel under severe data scarcity by mitigating rank deficiency, a hybrid parameterization that decouples the shape and volume terms of the log-likelihood function yields the superior accuracy–calibration balance as the class sample budget expands.

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Chapter 1

Introduction

1.1 Motivation

State-of-the-art (SOTA) object classification methods rely on deep learning models that are either trained end-to-end on the target application or initially trained on one task and subsequently fine-tuned for a different one. In both cases the resulting models are *monolithic*: a single network is trained on a large dataset and performs feature extraction and classification jointly. Such models are highly effective when the set of object classes is fixed and known at training or fine-tuning time. When new object classes must be incorporated, however, the entire model must be retrained or fine-tuned. This procedure requires significant computational resources, demands access to the original training dataset and, in many cases, leads to *catastrophic forgetting* of previously learned classes [6].

A natural way to integrate new classes while avoiding catastrophic forgetting is to formulate the problem as continual or class-incremental classification, in which the algorithm must learn to distinguish an ever-growing number of categories [7]. In this paradigm, a *task* corresponds to a distinct portion of the training experience, presented sequentially to the model. Instead of having access to the entire dataset at once, as in conventional supervised learning, the model receives data in sequential chunks [8]. Each task typically corresponds to a set of new classes that must be recognised. The central challenge is that when the model is updated task by task, it tends to overwrite previously acquired knowledge: monolithic models are structurally ill-suited for this scenario, and an alternative paradigm is therefore required.

This limitation has motivated the SOTA in continual classification to focus on balancing *stability* and *plasticity*. Stability refers to a model's ability to preserve previously learned knowledge; plasticity describes its capacity to adapt and incorporate new tasks [9]. Traditional approaches rarely optimise both properties simultaneously: when stability dominates, the model struggles to integrate new knowledge; when plasticity dominates, it loses access to past information. This trade-off is compounded by the limited capacity of neural networks to integrate new representations without interfering with old ones and by the need to continually adjust the network's structure to match the growing number of object classes. The

dominant view in the literature therefore treats continual learning as a problem of *parameter interference*: learning new classes requires updating network weights, which may overwrite previously acquired knowledge, and most existing methods attempt to control this process through regularisation, rehearsal buffers, or architectural adaptations. Although effective, these strategies share a common premise that this thesis sets out to challenge: that incremental learning must be achieved through continuous, careful optimisation of the parameters of the network responsible for feature extraction.

This premise diverges significantly from the organisational principles of biological vision. The human visual system relies on early cortical regions responsible for feature extraction that mature and stabilize during the initial postnatal weeks or months [10], providing highly stable feature representations that remain stationary throughout adulthood. However, despite this invariant sensory foundation, human lifelong learning persists. Higher-order cognitive systems seamlessly map these fixed perceptual representations onto newly encountered semantic categories, even if acquiring knowledge of a new object class can occasionally result in a slight, localized reduction in the ability to distinguish highly similar classes over time.

A directly analogous structural separation has recently become achievable in computer vision through large-scale visual foundation models. By leveraging encoders trained on massive, diverse datasets—such as DINOv2 [11]—these models produce highly discriminative representations that remain entirely stable once frozen. In this stationary feature space, *representation drift*—the gradual shifting of a network's internal features during sequential task updates, which serves as the primary driver of catastrophic forgetting—is eliminated by construction: the coordinates describing each class no longer change when a new task arrives. Mirroring the biological paradigm, class-incremental learning can thus be conceptually reframed: rather than protecting network parameters from destructive interference, the challenge becomes one of accurately modeling the distribution of new classes within a robust, fixed embedding space. This perspective motivates the present work: once a sufficiently discriminative and stable feature space is secured, the core challenge of continual learning shifts from *preventing forgetting* to *estimating probability densities*.

Building on this observation, this thesis proposes a continual-learning object-recognition system organized into two complementary, decoupled components. As illustrated in Figure 1.1, the first component is a *feature extractor* derived from a frozen, pre-trained foundation model. This encoder maps an input image to a static, highly discriminative feature vector $\vec{x}$ in a shared embedding space. Crucially, these representations must be sufficiently general to characterize arbitrary object classes, including those unseen during the foundation model's pre-training. The second component is a *statistical classifier* composed of a collection of independent, class-conditional Gaussian models $(G_1, \ldots, G_N)$. When a new class arrives, a corresponding Gaussian model is estimated and integrated without altering or accessing previously learned class models. At inference time, classification is performed by computing the posterior probability $p(c_k|\vec{x})$ for each class and assigning the input to the class that yields the maximum value, in direct analogy to a generative Bayes classifier. This modular, two-stage framework is developed in full mathematical detail in Chapter 3.

Gaussian class representations offer two key advantages that make them particularly well suited to this reframed problem. First, each class is modelled independently from the rest, so incorporating a new class amounts to estimating a small number of sufficient statistics

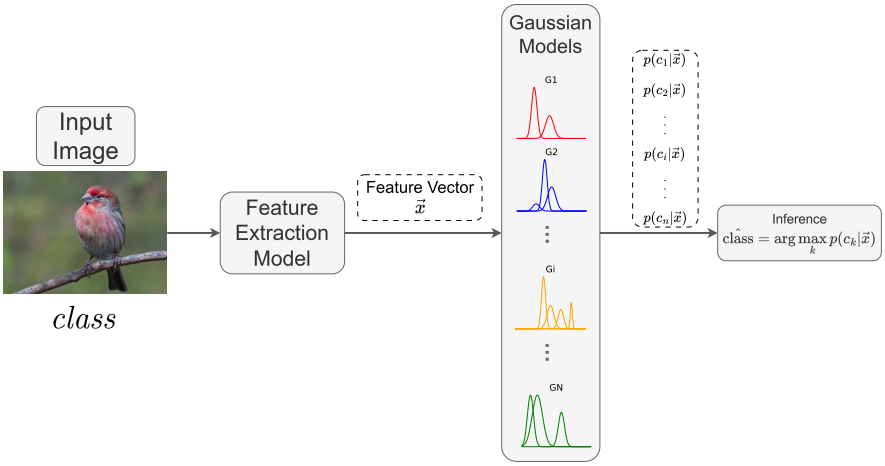


Figure 1.1: Conceptual overview of the proposed continual-learning framework. An input image is mapped by a frozen visual foundation model to a stable feature vector $\vec{x}$. Each object class c_k is modeled independently by a class-conditional Gaussian distribution (G_k). New classes are registered dynamically by estimating their unique parameters without modifying existing models. Inference selects the class that maximizes the posterior probability $p(c_k|\vec{x})$ under a Bayesian formulation. This schematic serves as an introductory conceptual illustration; Chapter 3 introduces a detailed architectural block diagram (Figure 3.1) that formalizes the recursive statistical update mechanisms and covariance estimators.

(mean and covariance) and does not require any modification of the previously stored class models; this yields incremental updates that are exact and gradient-free, and that produce zero catastrophic forgetting by construction. Second, Gaussian models are compact: they encode each class in a few parameters that can be estimated from a modest sample budget and refined as new samples become available, without any storage of the underlying images. This second property, however, is far from trivial in practice: because the dimensionality d of foundation-model embeddings (typically in the hundreds) is frequently comparable to, or larger than, the number of training samples n_c available per class, the empirical covariance matrix is structurally rank-deficient whenever $n_c < d$. This thesis studies this small-sample regime in detail, evaluating a full spectrum of covariance parameterisations—from diagonal to full covariance matrices, including shared and shrinkage-based estimators.

Because the entire framework rests on the assumption that class-conditional features are approximately Gaussian, this assumption cannot be taken for granted: it must be empirically validated rather than merely postulated. This thesis therefore complements its architectural proposal with a rigorous statistical characterisation of the DINOv2 embedding space, combining moment-based diagnostics (skewness and excess kurtosis) with formal hypothesis tests—the D'Agostino–Pearson omnibus test [12, 13] for univariate normality and a diagonal adaptation of the Henze–Zirkler test [14] for multivariate normality under severe rank deficiency. In parallel, the *Bhattacharyya distance* is used to quantify inter-class separability and to establish a principled theoretical ceiling on the accuracy that any Gaussian classifier can achieve in a given feature space. This separability analysis is not merely descriptive: it is used throughout this thesis to identify, explain, and visualise the specific pairs of classes—such as *boy*/*girl*, *bow-tie*/*scarf*, or *top-hat*/*cowboy-hat*—whose overlapping Gaussian distributions constitute the dominant source of classification error observed in this framework, as opposed

to any degradation of previously learned representations.

1.2 Hypothesis

We hypothesize that the combination of an independent class-conditional Gaussian model and a frozen pre-trained feature extractor provides a robust framework for continual object classification. By enabling the addition of new classes without full-model retraining, this design eliminates catastrophic forgetting while maintaining a simple, compact architecture.

1.3 General Objective

The main objective of this thesis is to develop and to validate a continual learning framework for object classification using pre-trained foundation models and independent class-conditional Gaussian models as classifiers, with the purpose of enabling the incremental incorporation of new classes while preventing catastrophic forgetting

1.4 Specific Objectives

For the purpose of fulfilling the general objective and verifying the hypothesis of this work, the following specific objectives are defined:

- To develop, evaluate, and characterize the practical behavior of the proposed Gaussian classifier against state-of-the-art baselines, assessing its sensitivity to covariance-matrix estimators, Mixture-of-Gaussian variants, and upstream feature extractors, while analyzing its sample efficiency, probabilistic calibration, and computational requirements.
- To statistically validate the underlying assumptions of the framework by analyzing the dataset representations within the feature extractor embedding space, specifically assessing the Gaussianity of class-conditional feature distributions, evaluating their spectral structure, and quantifying inter-class separability to identify the sources of residual classification error.

1.5 Contributions

The principal contributions of this thesis are the following:

1. A gradient-free class-incremental learning framework that pairs a frozen DINOv2 backbone with independent, closed-form class-conditional Gaussian models, eliminating

catastrophic forgetting by construction rather than by mitigation, together with a systematic evaluation of five covariance parameterisations (diagonal, full, hybrid, shared, and shrinkage-based) and their mixture-of-Gaussian extensions.

2. A rigorous, multi-layered statistical validation of the Gaussianity assumption underlying the framework—combining moment-based diagnostics, the D'Agostino–Pearson omnibus test, and a diagonal adaptation of the Henze–Zirkler test for both univariate and multivariate normality under severe rank deficiency—that had not previously been carried out for foundation-model embedding spaces at this level of detail.
3. A quantitative link between inter-class separability, measured via the pairwise Bhattacharyya distance, and the specific, named class pairs that account for the residual classification error of the proposed classifier, reframing that error as a property of the embedding space rather than of the classifier itself.

1.6 Thesis Structure

The remainder of the document is organised as follows.

Chapter 2 – Theoretical Framework and Related Work. Reviews the machine-learning and computer-vision background needed to situate this work, from Bayesian classification and Gaussian mixture models to covariance estimation in high-dimensional, small-sample regimes, before surveying the continual-learning literature—rehearsal-, regularisation-, architecture-, and prompt-based methods, as well as the classifier-adaptation and class-prototype approaches most closely related to the framework proposed here.

Chapter 3 – Modelling Class Densities in a Frozen Feature Space. Presents the proposed framework in full mathematical detail: the frozen-encoder-plus-Gaussian-classifier architecture, the datasets and evaluation protocol, the family of covariance estimators and mixture-of-Gaussian variants, the confidence and calibration metrics, the incremental class-inclusion protocol, the inter-class Bhattacharyya distance, the backbone-sensitivity study, and the statistical tests used to validate the Gaussianity assumption.

Chapter 4 – Experimental Results and Analysis. Reports and interprets the experimental campaign, following the same order in which the corresponding topics are introduced in Chapter 3: covariance-estimator and mixture-of-Gaussian performance, calibration, incremental class inclusion, inter-class separability, backbone sensitivity, spectral structure of the embedding space, and the empirical validation of the Gaussianity assumption, closing with a synthesis of practical recommendations.

Chapter 5 – Conclusion. Summarises the central findings of the thesis at the level of the framework as a whole and outlines directions for future work that follow directly from the limitations identified in Chapter 4.

Chapter 2

Theoretical Framework and Related Work

This chapter establishes the conceptual and mathematical foundations required to contextualize this research, transitioning from broad paradigms in Artificial Intelligence to the specific statistical mechanics and adaptive learning strategies underpinning our methodology. We begin by reviewing the fundamental principles of Machine Learning, specifically within the realms of image processing and object classification, before providing a rigorous mathematical treatment of Bayesian Inference, Mixture of Gaussians (MoG) and Gaussian Analysis, including the moment-based and hypothesis-testing tools used throughout this thesis to assess Gaussianity. We then examine the theoretical foundations of covariance estimation in high-dimensional, small-sample regimes—the curse of dimensionality, covariance degeneracy, and the two families of regularisation strategies, spectral conditioning and structured shrinkage, that motivate the specific estimators (diagonal, full, hybrid, shared, OAS, and Ledoit–Wolf) developed in full in Chapter 3. Finally, we address the challenge of "catastrophic forgetting" by introducing the Continual Learning (CL) paradigm, describing the datasets utilized and evaluating the state-of-the-art mitigation strategies that provide a performance baseline for the proposed work.

2.1 Artificial Intelligence and Machine Learning

Artificial Intelligence (AI) is a broad field of computer science dedicated to creating systems capable of performing tasks that typically require human intelligence, such as learning, reasoning, perception, and language understanding [15]. Within this domain, Machine Learning (ML) focuses on developing algorithms that allow computers to learn patterns from data without explicit programming. ML systems are generally categorized into three paradigms: Supervised Learning, Unsupervised Learning, and Reinforcement Learning [16].

First, in **Supervised Learning**, the model learns a mapping from input variables $\mathbf{x}$ to output variables y based on a labeled dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$. The objective is to minimize a loss function that penalizes the discrepancy between the predicted output and the ground

truth. This paradigm is typically divided into Classification, where the output is a discrete category, and Regression, where the output is a continuous value.

Second, **Unsupervised Learning** deals with unlabeled data $\mathcal{D} = \{\mathbf{x}_i\}_{i=1}^{N}$. Instead of predicting a target, the goal is to discover the underlying structure or distribution of the data. Common tasks include Clustering, which groups similar data points (e.g., K-Means or Gaussian Mixture Models), and Dimensionality Reduction, which projects high-dimensional data into a lower-dimensional space while preserving essential information (e.g., PCA or t-SNE).

Finally, **Reinforcement Learning** is based on an agent interacting with an environment to maximize a cumulative reward. Unlike supervised learning, the agent is not told which actions to take but must discover them through trial and error. This process is often modeled as a Markov Decision Process (MDP) [17], where the agent transitions between states based on actions, receiving feedback in the form of numerical rewards. The focus is on finding an optimal policy that balances exploration (trying new actions) and exploitation (using known high-reward actions).

2.1.1 Deep Learning

In traditional Machine Learning, the feature extraction step is a manual process where a human expert must identify and define the most relevant characteristics (such as the shape or edges of the car) before passing them to a classification algorithm. In contrast, as shown in figure 2.1, Deep Learning utilizes neural networks to automate this process; the model performs both feature extraction and classification simultaneously, learning to identify complex patterns directly from the input data without human intervention.

Structurally, this shift allows Deep Learning to handle much more complex, high-dimensional data, though it typically requires significantly larger datasets and more computational power to be effective. While traditional Machine Learning depends on the quality of hand-crafted features, Deep Learning architectures—like multi-layered neural networks—are designed to build hierarchical representations of data, progressively moving from low-level pixels to high-level object concepts.

2.1.2 Computer Vision and Vision Transformers

Computer Vision is the interdisciplinary subfield of AI that enables computers to gain high-level understanding from digital images or videos, effectively automating tasks performed by the human visual system [18]. While Convolutional Neural Networks (CNNs) [19] historically dominated this field, the Transformer architecture—originally developed for Natural Language Processing—has revolutionized visual recognition through the self-attention mechanism [20].

Vision Transformers (ViT) adapt this architecture by treating an image as a sequence of discrete tokens. As illustrated in Figure 2.2, the process follows a specific pipeline:

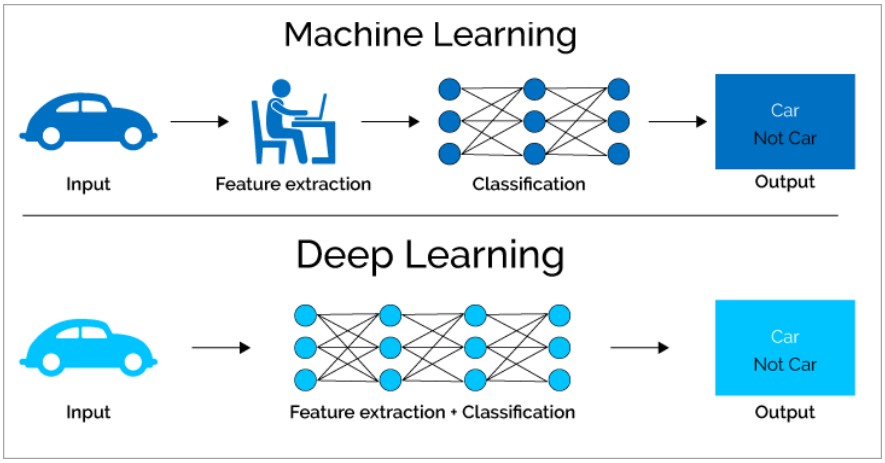


Figure 2.1: Machine Learning and Deep Learning Strategies. Image taken from [1].

- **Patch Partitioning:** The input image is decomposed into fixed-size patches.
- **Linear Projection:** These patches are flattened and mapped to an embedding space.
- **Positional Encoding:** Since Transformers lack inherent knowledge of spatial structure, learnable position embeddings are added to the patch embeddings.
- **Transformer Encoder:** The resulting sequence is processed through multiple layers of multi-head self-attention and MLP blocks to capture global dependencies.

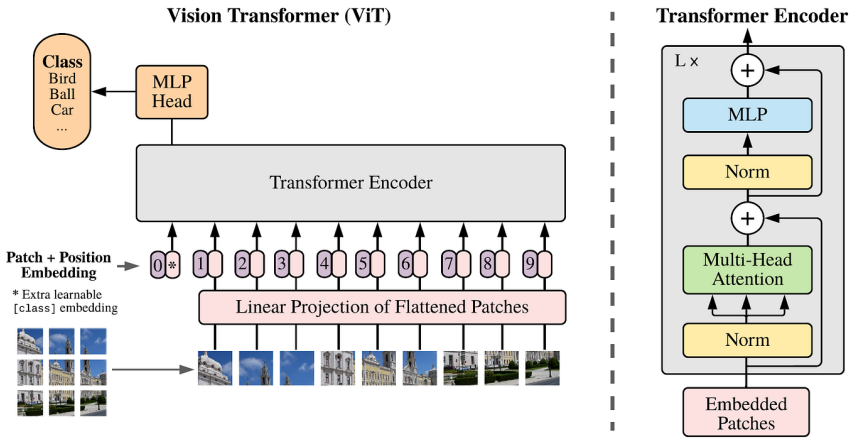


Figure 2.2: Vision Transformer (ViT) architecture overview. The model processes an image as a sequence of patches through a standard Transformer encoder. Image taken from [2].

Object Classification and Representative Power

The primary objective of the original ViT was *Object Classification*: the fundamental task of assigning a single categorical label to an entire input image. This task serves as the primary benchmark for evaluating the representative power of visual models [21]. By processing the global sequence of patches, ViTs have demonstrated an ability to meet or exceed CNN performance, particularly when pre-trained on the large-scale datasets described below.

Self-Supervised Advancements: DINOv2

While most datasets rely on explicit human annotations, recent advancements have shifted toward foundational models trained via self-supervised learning (SSL). A prominent representative of this paradigm is **DINOv2** [11], which produces highly robust, multipurpose visual features that exhibit exceptional "out-of-the-box" performance across both image-level and pixel-level tasks.

The underlying architecture of DINOv2 relies on a joint **Student-Teacher self-distillation** framework, building upon DINO [22] and iBOT [23]. It operates with two parallel networks of identical configuration: a *Student* network g_{θ_s} and a *Teacher* network g_{θ_t}, whose weights θ_t are updated as an Exponential Moving Average (EMA) of the student's weights:

$$\theta_t \leftarrow \lambda\theta_t + (1-\lambda)\theta_s \tag{2.1}$$

where $\lambda \in [0, 1]$ is a smoothing parameter. During training, the framework optimizes two complementary objectives to learn both global and fine-grained spatial representations:

- **Image-Level Loss (DINO):** The student is fed with both global and local image crops, while the teacher processes only global views. The student is trained to match the teacher's output distribution over a set of virtual prototype vectors, stabilized using an online *Sinkhorn-Knopp centering* algorithm to prevent representation collapse.
- **Patch-Level Loss (iBOT):** To force the model to capture dense, localized features, random patches of the student's global views are masked out. The student must reconstruct the corresponding local patch features produced by the teacher, which receives the fully unmasked global view.
- **KoLeo Regularization:** To ensure that the resulting embeddings are distributed uniformly across the feature space and do not collapse into a narrow cone, DINOv2 incorporates the Kozachenko-Leonenko (KoLeo) differential entropy estimator:

$$\mathcal{L}_{\text{KoLeo}} = -\frac{1}{n}\sum_{i=1}^{n} \log\left(\min_{j\neq i} \|\vec{x}_i - \vec{x}_j\|_2\right) \tag{2.2}$$

  where $\vec{x}_i$ represents the ℓ_2-normalized feature vector of the i-th image sample in a batch of size n.

2.1.3 Image Datasets and Benchmarks

To rigorously test the robustness and generalization of architectures like ViT, this work utilizes several standardized datasets that vary in scale, stylistic diversity, and granularity.

The ImageNet Family

ImageNet [24] is a cornerstone of computer vision, providing over 1.2 million images across 1,000 classes. Its hierarchical structure, based on WordNet, makes it the standard for large-scale classification. To further push the boundaries of model performance, two variants are often employed:

- **ImageNet-21K (and ImageNet-21k-P):** An expansive version containing 14 million images across 21,000 classes [25]. The "P" version is a cleaned subset designed to make pre-training for models like ViT more accessible.
- **ImageNet-R (Renditions):** A dataset of 30,000 images covering 200 classes in non-photographic styles (e.g., art, sketches). It tests a model's robustness to out-of-distribution data [26].

Fine-Grained and Diverse Benchmarks

Beyond large-scale pre-training, fine-grained classification is evaluated using:

- **CIFAR-100:** Consists of 60,000 low-resolution (32×32) images [27]. It is unique for its two-level hierarchy, where 100 "fine" classes are nested within 20 "coarse" superclasses, allowing for the evaluation of both specific and broad categorization.
- **Caltech-256:** Provides 30,607 images across 256 categories [28]. It offers greater category diversity and a higher minimum number of images per class than its predecessors, serving as a robust benchmark for generalisation. In this thesis we use the first 250 categories (approximately 28,100 images) so that the resulting class-incremental protocol can be divided into 10 balanced tasks of 25 classes each (see Chapter 3).

2.2 Foundations of Probabilistic Modeling

The convergence of statistical learning theory and high-dimensional data analysis has necessitated the development of robust mathematical frameworks to handle the dual challenges of uncertainty quantification and dimensionality reduction. As modern neural architectures, such as the DINOv2 Vision Transformer, extract increasingly complex representations from datasets like ImageNet-R, the subsequent classification and density estimation layers must be grounded in rigorous probabilistic foundations.

2.2.1 Bayes Classification

At its core, Bayes classification is rooted in **Bayes' Theorem**, which provides a mathematical framework for updating the probability of a hypothesis based on observed evidence. For a

classification task, we seek the posterior probability $p(y = c \mid \vec{x})$—the probability that an observation belongs to class c given its observed feature vector $\vec{x}$.

Bayes' Theorem is defined as:

$$p(y = c \mid \vec{x}) = \frac{p(\vec{x} \mid y = c)\, p(y = c)}{p(\vec{x})} \tag{2.3}$$

where:

- $p(y = c \mid \vec{x})$ is the **posterior** probability.
- $p(\vec{x} \mid y = c)$ is the **likelihood** of the data given the class.
- $p(y = c)$ is the **prior** probability of the class.
- $p(\vec{x})$ is the **evidence** or marginal likelihood, acting as a normalization constant.

The objective of a Bayes classifier is to minimize the Bayes risk; in the simplest case, this corresponds to assigning the observation to the class that maximizes the posterior probability.

2.2.2 Likelihood-Based Classification

Likelihood inference provides a coherent framework for updating the probability of a hypothesis as more evidence or information becomes available. In contrast to frequentist approaches that treat parameters as fixed but unknown constants, the Bayesian paradigm treats parameters as random variables characterized by probability distributions.

The connection between Bayes' Theorem and likelihood-based classification lies in the **Maximum A Posteriori (MAP)** decision rule. Since the evidence $p(\vec{x})$ is constant across all classes for a given observation, the classification decision depends solely on the product of the likelihood and the prior:

$$p(y = c \mid \vec{x}) \propto p(\vec{x} \mid y = c)\, p(y = c). \tag{2.4}$$

A generative modeling framework can be adopted and class-conditional Gaussian feature distributions can be assumed:

$$p(\vec{x} \mid y = c) = \mathcal{N}(\vec{x} \mid \boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c), \tag{2.5}$$

where $\vec{x} \in \mathbb{R}^d$ represents the feature vector, $\boldsymbol{\mu}_c$ is the class mean vector, and $\boldsymbol{\Sigma}_c \succ 0$ is the covariance matrix.

Classification can be performed via MAP inference by maximizing the log-posterior:

$$\hat{y} = \arg\max_{c \in \mathcal{Y}} \left[\log p(\vec{x} \mid y = c) + \log \pi_c\right], \tag{2.6}$$

where $\pi_c = p(y = c)$ denotes the class prior.

Substituting the multivariate Gaussian density into the MAP rule, the decision rule becomes equivalent to:

$$\hat{y} = \arg\min_{c} \left[(\vec{x} - \boldsymbol{\mu}_c)^T \boldsymbol{\Sigma}_c^{-1} (\vec{x} - \boldsymbol{\mu}_c) + \log |\boldsymbol{\Sigma}_c| - 2 \log \pi_c\right]. \tag{2.7}$$

Mixture of Gaussians (MoG)

A Mixture of Gaussians models a complex probability density as a convex combination of K distinct Gaussian components. This approach is fundamental for clustering and density estimation in multi-modal distributions. Therefore, to capture multimodal feature distributions, the class-conditional model can be extended to a mixture:

$$p(\vec{x} \mid y = c) = \sum_{k=1}^{K_c} \alpha_{c,k} \mathcal{N}(\vec{x} \mid \boldsymbol{\mu}_{c,k}, \boldsymbol{\Sigma}_{c,k}), \tag{2.8}$$

where $\alpha_{c,k} \geq 0$ and $\sum_k \alpha_{c,k} = 1$.

The MAP decision rule becomes

$$\hat{y} = \arg\max_c \left[\log \left(\sum_{k=1}^{K_c} \alpha_{c,k} \mathcal{N}(\vec{x} \mid \boldsymbol{\mu}_{c,k}, \boldsymbol{\Sigma}_{c,k}) \right) + \log \pi_c \right]. \tag{2.9}$$

2.2.3 Gaussianity Assumptions and Analysis

In the context of generative modeling for foundation feature spaces, the validity of the Bayes classifier depends on how well the class-conditional Gaussian distributions approximate the true data distribution. The following analytical framework establishes the statistical basis for these assumptions.

Principal Component Analysis

Principal Component Analysis (PCA) is a linear dimensionality-reduction technique that finds an orthonormal basis for the data by diagonalising the empirical covariance matrix [29]. Given a centred data matrix $\mathbf{X} \in \mathbb{R}^{N \times d}$ (zero-mean columns), the empirical covariance is

$$\mathbf{S} = \frac{1}{N-1} \mathbf{X}^T \mathbf{X}. \tag{2.10}$$

Its eigendecomposition $\mathbf{S} = \mathbf{P} \boldsymbol{\Lambda} \mathbf{P}^T$ yields orthonormal eigenvectors $\mathbf{P} = [\mathbf{p}_1, \ldots, \mathbf{p}_d]$ sorted so that the corresponding eigenvalues satisfy $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0$. Each eigenvector $\mathbf{p}_i$ is a *principal component* (PC), and λ_i equals the variance of the data projected onto $\mathbf{p}_i$.

Dimensionality reduction. Retaining only the leading k eigenvectors defines the projection matrix $\mathbf{P}_k = [\mathbf{p}_1, \ldots, \mathbf{p}_k] \in \mathbb{R}^{d \times k}$. A new observation $\vec{x} \in \mathbb{R}^d$ is mapped to a k-dimensional representation via

$$\tilde{\vec{x}} = \mathbf{P}_k^T(\vec{x} - \boldsymbol{\mu}), \tag{2.11}$$

where $\boldsymbol{\mu}$ is the global mean estimated from the training corpus. The resulting coordinates $\tilde{\vec{x}}$ are decorrelated: $\mathrm{Cov}(\tilde{\vec{x}}) = \mathrm{diag}(\lambda_1, \ldots, \lambda_k)$. The fraction of total variance preserved by the projection is

$$R(k) = \frac{\sum_{i=1}^{k} \lambda_i}{\sum_{i=1}^{d} \lambda_i}. \tag{2.12}$$

Optimality and geometric interpretation. Among all rank-k linear projections, PCA minimises the mean-squared reconstruction error and maximises the variance of the projected data (Eckart–Young theorem [30]). Geometrically, the projected subspace aligns with the axes of the data ellipsoid along which the observations spread most.

Application to high-dimensional embeddings. When the original dimension d is large (e.g. $d = 768$ for DINOv2 ViT-B/14) but the data occupy a low-dimensional subspace—as evidenced by rapid eigenvalue decay—reducing to $k \ll d$ dimensions before downstream classification offers two complementary benefits: (i) it concentrates the discriminative signal in fewer coordinates, potentially improving the accuracy of subsequent covariance estimators; and (ii) it reduces computational cost by a factor of $\mathcal{O}((d/k)^2)$ for quadratic (Mahalanobis) classifiers. The trade-off is that components beyond k are discarded, which may remove residual discriminative information if the spectrum does not decay quickly enough.

Spectral Decay and Effective Dimensionality

The practical validity of high-dimensional Gaussian modeling is further supported by examining the spectral structure of the feature space through Principal Component Analysis (PCA). The covariance of the class-mean-centred embeddings can be decomposed as $\boldsymbol{\Sigma} = \mathbf{P}\boldsymbol{\Lambda}\mathbf{P}^T$, where the diagonal of $\boldsymbol{\Lambda}$ contains the eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0$ and $\mathbf{P}$ holds the corresponding orthonormal eigenvectors.

Variance concentration. If the eigenvalue spectrum decays steeply—most variance is explained by the first $k \ll d$ components—then the effective dimensionality of the feature distribution is much lower than the nominal d. Concretely, the fraction of variance explained by the leading k components is $\sum_{i=1}^{k} \lambda_i \,/\, \sum_{i=1}^{d} \lambda_i$.

Univariate Gaussianity

A second layer of theoretical justification involves examining the univariate properties of individual feature components.

Kurtosis This statistic measures the "tailedness" and peakedness of a probability distribution relative to a normal distribution [3]. As illustrated in Figure 2.3, a perfectly Gaussian distribution has an excess kurtosis of zero (mesokurtic). Distributions with high kurtosis (leptokurtic, $Kurtosis > 0$) exhibit sharper peaks and heavier tails, indicating a higher concentration of data around the mean and a greater likelihood of extreme outliers. Conversely, low kurtosis (platykurtic, $Kurtosis < 0$) indicates a flatter peak and thinner tails. Empirical evidence suggests that for many foundation model embeddings, kurtosis values typically fall within the range of $[-1, 1]$, justifying the Gaussian approximation.

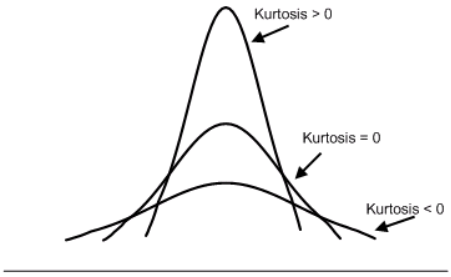


Figure 2.3: Visual representation of kurtosis levels: leptokurtic (> 0), mesokurtic ($= 0$) and platykurtic (< 0) distributions [3].

Skewness This measure quantifies the asymmetry of a distribution around its mean. As shown in Figure 2.4, a symmetric distribution (center) is a fundamental assumption of the Gaussian model, implying that the mean $\boldsymbol{\mu}_c$, median, and mode coincide, providing an adequate representation of the class prototype within the embedding space. A positive skew (right) indicates a distribution with a long tail extending toward higher values, while a negative skew (left) indicates a long tail extending toward lower values. Significant skewness would suggest that the class-conditional mean is no longer a central representation of the feature distribution.

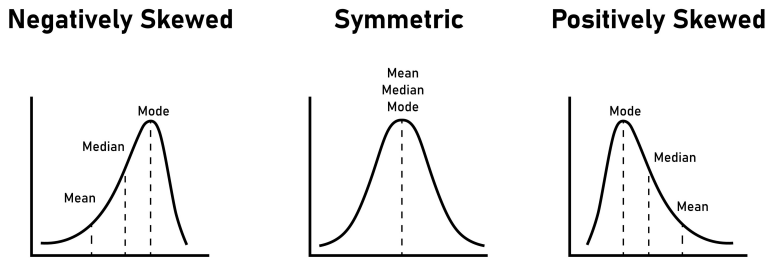


Figure 2.4: Comparison of negatively-skewed, symmetric and positively-skewed distributions [4].

Multivariate Gaussianity

Per-dimension normality is necessary but not sufficient for the joint d-dimensional distribution to be Gaussian. The Henze–Zirkler (HZ) test [14] is an omnibus multivariate diagnostic that addresses this gap by comparing the empirical characteristic function of the data against the theoretical characteristic function of a multivariate normal distribution.

Given n centred samples $\vec{z}_i = \vec{x}_i - \hat{\boldsymbol{\mu}}_c$ and a smoothing bandwidth $b > 0$, the test statistic is built from three terms evaluated using the squared Mahalanobis distances $d_M^2(\cdot, \cdot)$ induced by the class covariance $\boldsymbol{\Sigma}_c$:

$$A = \frac{1}{n^2} \sum_{j=1}^{n} \sum_{k=1}^{n} e^{-\frac{b^2}{2} d_M^2(\vec{z}_j, \vec{z}_k)}, \tag{2.13}$$

$$B = \frac{2}{n} (1 + b^2)^{-d/2} \sum_{j=1}^{n} e^{-\frac{b^2}{2(1+b^2)} d_M^2(\vec{z}_j, \vec{0})}, \tag{2.14}$$

$$C = (1 + 2b^2)^{-d/2}. \tag{2.15}$$

Term A measures the average pairwise kernel similarity of the observed samples; term B quantifies how consistently the empirical density decays away from the class centroid compared to a Gaussian; and term C is the asymptotic self-similarity constant of a true d-dimensional Gaussian. The final statistic is:

$$T_{n,b} = n \cdot [A - B + C]\,. \tag{2.16}$$

Under the null hypothesis of multivariate normality, $T_{n,b}$ converges to a fixed value close to 1; departures above 1 indicate that the data is more structured (tighter clusters or heavier tails) than a Gaussian. The bandwidth b controls the resolution of the characteristic function comparison and is set optimally as a function of n and d.

2.2.4 Covariance Estimation in High-Dimensional Spaces

A fundamental challenge in multivariate statistical analysis and pattern recognition is the estimation of the covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{d \times d}$ from a finite set of n observations. In classical low-dimensional regimes, where the sample size n is much larger than the feature dimension d ($n \gg d$), the empirical Sample Covariance Matrix (SCM) acts as a reliable, unbiased estimator. However, modern engineering applications—such as analyzing high-dimensional deep feature representations—frequently operate in the "high-dimension, small-sample-size" (HDSS) regime, where d is of the order of hundreds or thousands, while n remains strictly limited.

Under these conditions, classical multivariate analysis suffers severely from the *curse of dimensionality*, leading to critical theoretical and numerical breakdowns in statistical modeling.

The Curse of Dimensionality and Covariance Degeneracy

When the dimensionality d scales while the sample size n remains fixed or grows at a slower rate, several geometric and algebraic anomalies emerge:

- **Singularity and Rank Deficiency:** Structurally, the SCM is calculated as the sum of n outer products of centered data vectors. Consequently, the maximum rank of the SCM is bounded by $\min(n-1, d)$. In the typical HDSS scenario where $n < d$, the SCM has at least $d-n+1$ zero eigenvalues. The resulting matrix is singular (rank-deficient) and cannot be inverted.
- **Loss of Negative Log-Likelihood and Distance Metrics:** Many fundamental statistical classifiers and density estimators—such as Gaussian Discriminant Analysis (GDA) or Mahalanobis distance formulations—depend heavily on the precision matrix (the inverse covariance, $\mathbf{\Sigma}^{-1}$) and the log-determinant $\ln|\mathbf{\Sigma}|$. When the estimated covariance is singular, the log-determinant diverges to $-\infty$, and the precision matrix is undefined, rendering density estimation mathematically impossible.
- **Eigenvalue Distortion (Spreading Phenomenon):** Even in moderately high dimensions where $n \geq d$ and the SCM is technically invertible, the sample eigenvalues do not converge to their true counterparts. As proved in random matrix theory (e.g., the Marčenko-Pastur law [31]), high-dimensional sample eigenvalues exhibit severe bias: the largest eigenvalues are systematically overestimated, while the smallest eigenvalues are underestimated. This artificial dispersion distorts the perceived geometric structure of the data.

Theoretical Foundations of Regularization

To overcome the numerical and structural instability of the empirical SCM in high dimensions, the theoretical framework must transition from pure maximum likelihood estimation toward *regularization*. The conceptual objective is to trade a manageable amount of bias for a drastic reduction in estimator variance.

Conceptually, these regularized estimation strategies fall into two primary paradigms:

1. **Spectral Regularization and Conditioning:** These approaches directly manipulate the empirical eigenspectrum. By placing a lower bound on near-zero eigenvalues or adding a perturbation matrix (e.g., diagonal loading), they artificially bound the condition number of the matrix. This guarantees strict positive definiteness ($\mathbf{\Sigma} \succ 0$), ensuring that the precision matrix remains numerically stable during inversion.
2. **Structured Shrinkage:** Shrinkage theory, pioneered by Stein's loss formulations, constructs an estimator as a convex combination of the high-variance SCM and a low-variance structured "target" matrix (such as the identity matrix or a diagonal variance matrix). By shrinking the empirical estimator toward a well-behaved target, these methods systematically pull the distorted, spread-out eigenvalues back toward their true mean, minimizing the expected mean squared error.

2.3 Continual Learning

Continual Learning (CL), also known as lifelong learning, constitutes the core focus of this work. It represents a machine learning paradigm designed to handle the progressive acquisition of knowledge from sequences of tasks or data streams over time. As illustrated in Figure 2.5, a **task** in this context is defined as a specific learning experience or dataset –such as Task A, B, or C. Each task is potentially characterized by unique data distributions or objectives, as symbolized by the varying shapes (e.g., circles vs. hearts) above the task labels.

Unlike conventional training approaches that rely on a fixed dataset and a stationary environment, CL emphasizes the arrival of these tasks at distinct intervals along the **time** axis. This sequential arrival requires the model to be inherently non-stationary, adapting its internal representations as environments evolve. The primary challenge in this paradigm lies in the ongoing integration of new information from a current task (e.g., Task B) while simultaneously preserving the knowledge previously acquired from Task A.

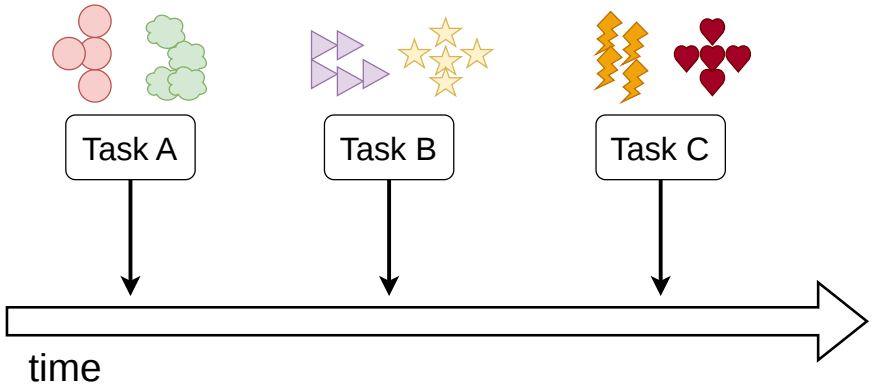


Figure 2.5: A sequential task arrival model for Continual Learning, depicting the transition from Task A to Task C over time, where each task represents a unique set of features or data distributions.

Without specific architectural or algorithmic safeguards, this sequential learning process inevitably triggers **catastrophic forgetting** [32]. This phenomenon occurs when the assimilation of new knowledge leads to the degradation or total loss of prior learning. As illustrated in Figure 2.6, the dilemma of forgetting becomes apparent when a *New Object Class* is introduced to an existing classifier, forcing a choice between two traditional but flawed strategies:

- **Fine-Tuning:** Updating the model using only the current task data is computationally efficient but leads directly to catastrophic forgetting [32]. Because the optimization process focuses solely on the new objective, the internal representations associated with previously learned classes are overwritten, causing the model to lose its original predictive capabilities.
- **Retraining from Scratch:** While joint training on the full historical dataset ($\mathcal{D}_{old} \cup \mathcal{D}_{new}$) technically prevents forgetting, it faces two significant hurdles. First is the issue

of **Data Privacy and Availability**: in many real-world scenarios, the original data $\mathcal{D}_{old}$ is often deleted or inaccessible due to privacy regulations. Second is the **Computational Bottleneck**: as the system's lifespan increases, the cumulative dataset grows, making the energy and time required to retrain the model unsustainable in continuously evolving environments.

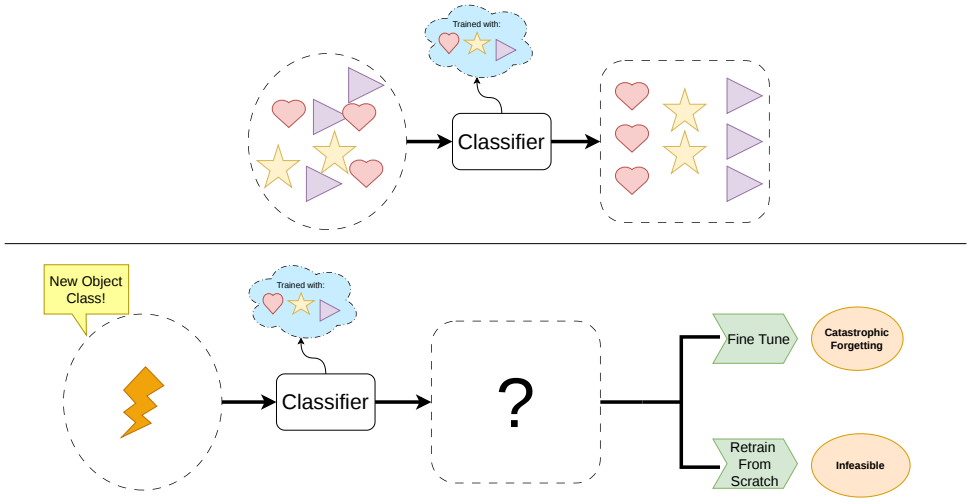


Figure 2.6: Standard Object Classifier comparison. The upper path shows standard inference, while the lower path illustrates the dilemma when a new class is introduced: the trade-off between catastrophic forgetting via fine-tuning and the infeasibility of retraining from scratch.

2.3.1 Elasticity and Plasticity

A central concept in continual learning is the balance between elasticity and plasticity. **Plasticity** refers to the ability of a model to adapt rapidly to new information, ensuring that it can effectively learn novel tasks. **Elasticity** (or stability), in contrast, emphasizes preserving the parameters and representations that are crucial for retaining old knowledge [9]. Achieving an appropriate equilibrium between these two properties is known as the **stability–plasticity dilemma** [32].

2.3.2 Main Paradigms

As shown in Figure 2.7, two principal paradigms define how continual learning problems are formulated. **Task-incremental learning (Task-IL)** assumes that the task identity is available during inference, allowing for task-specific components to be isolated [33]. **Class-incremental learning (Class-IL)**, on the other hand, does not provide the task identity at test time, requiring the model to distinguish between all classes seen across all tasks simultaneously [5].

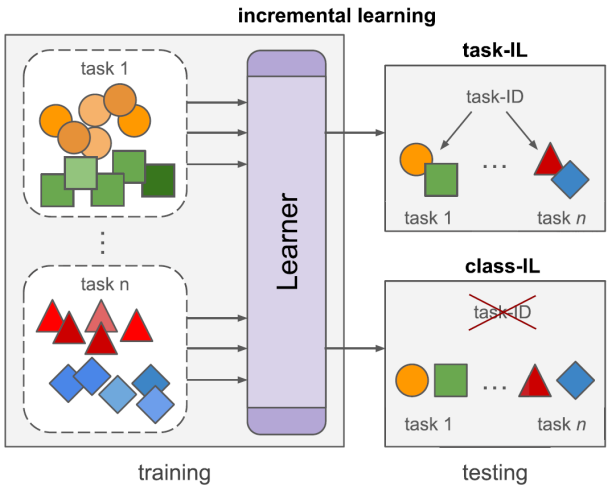


Figure 2.7: Class and Task Incremental Learning. Image taken from [5].

2.3.3 Mitigation Strategies

To address the stability-plasticity dilemma, we categorize existing CL strategies into four main families: rehearsal-based, regularization-based, architecture-based, and representation-based (specifically prompt-based) methods. Additionally, emerging paradigms focus on classifier adaptation over frozen backbones.

Rehearsal Methods

To preserve past performance, these methods reuse samples from previously learned tasks while training on the current task.

- **Experience Replay (ER)** [34]: Maintains a small episodic memory buffer containing a sample of old, representative examples to be interleaved with new data.
- **Average Gradient Episodic Memory (A-GEM)** [35]: Improves efficiency by using the buffer to constrain the direction of parameter updates, ensuring they do not interfere with past tasks.
- **Dark Experience Replay (DER)** [36]: Enhances ER by incorporating knowledge distillation on the logits of replayed examples to preserve the "dark knowledge" of the model.
- **Generative Replay** [37]: A non-data-storing method where a generator is trained to produce synthetic features or samples of previously learned classes, avoiding the privacy concerns of raw data storage.

Regularization Methods

These methods limit updates to parameters identified as important for prior learning by introducing penalty terms in the loss function.

- **Elastic Weight Consolidation (EWC)** [38]: Identifies critical parameters using the Fisher Information Matrix and restricts their movement during new task training.
- **Synaptic Intelligence (SI)** [39]: Calculates importance weights online by accumulating their contribution to the reduction of the loss.
- **Learning Without Forgetting (LwF)** [40]: A functional regularization method that uses knowledge distillation to ensure the model's outputs for new data remain consistent with the previous model's predictions.
- **Adaptive Plasticity Improvement (API)** [9]: Enforces an update direction orthogonal to the important subspaces of previous tasks using Dual Gradient Projection Memory (Dual-GPM).

Architecture-based Methods

These methods mitigate forgetting by modifying the network architecture to increase or reorganize its capacity.

- **Parameter Isolation** [41]: Identifies and freezes parameters important for previous tasks—often through pruning or gradient masking—so only a subset of weights is updated for the current task.
- **Dynamic Expansion**: Expands the model architecture as new tasks arrive. For instance, DER [42] uses an expandable super-feature extractor followed by pruning, while DEN [43] expands the network dynamically only when the current capacity is insufficient.

Representation and Prompt-based Methods

The emergence of foundation models (e.g., ViTs) has enabled parameter-efficient fine-tuning (PEFT) where the backbone remains frozen and learning is contained within small, task-specific parameters ("prompts").

- **Learning to Prompt (L2P)** [44]: Uses a query-based prompt pool where prompts are selected based on the similarity between input embeddings and learned keys.
- **Dual Prompt** [45]: Refines L2P by separating prompt-based representations into complementary components: fixed G-prompts (General) and task-specific E-prompts (Expert).

- **CODA-Prompt** [46]: Introduces a fully differentiable scheme where the prompt is a weighted sum of components dynamically calculated via attention mechanisms.
- **Advanced Prompting**: Recent extensions adapt prompt engineering to parameter-efficient and distributed settings. Specifically, **InFLoRA** [47] and **DOLFIN** [48] apply low-rank adaptation and distributed learning, while **APT** [49] injects key-value tokens directly into transformer layers to minimize inference overhead.

Classifier Adaptation on Frozen Representations

A complementary line of work investigates classifier adaptation on top of frozen feature extractors, remaining fundamentally discriminative or metric-based without backpropagating through the backbone.

- **SLCA** [50]: Proposes a slow learner with classifier alignment for continual learning on pre-trained models, enabling efficient incremental learning without full-network fine-tuning.
- **ACIL** [51]: Introduces analytic class-incremental learning with closed-form updates for linear classifiers, providing absolute memorisation and privacy protection without requiring exemplar storage.
- **FeCAM** [52]: Exploits the heterogeneity of class distributions in exemplar-free continual learning by performing classification using a Mahalanobis distance metric to capture complex feature distributions.

Class Prototype and Analytic Classifier Methods

A closely related family of methods represents each class through distances to class prototypes (first-order information), sometimes complemented by second-order statistics.

- **SimpleCIL** [53]: Performs first-session adaptation by applying PEFT to the backbone only during the initial task, after which class prototypes are computed as the mean embedding of the samples belonging to each class and kept fixed for the remainder of the stream.
- **RanPAC** [54]: Improves upon prototype-based schemes by applying a random projection followed by a non-linear activation. This lifts features into a higher-dimensional space where classes are more linearly separable, accumulating second-order information through a shared, incrementally updated Gram matrix.
- **FeTrIL** [55]: Stores only the centroid representation of each past class. When a new task arrives, these centroids are used to translate the new class's features into pseudo-features that emulate past distributions, which are then used to train a linear classifier incrementally.

- **FeCAM** [52]: Explores Mahalanobis distances instead of typical Euclidean distances. It applies a Tukey power-ladder transformation to correct feature skewness before normalising each class covariance so that its diagonal entries equal one, making Mahalanobis distances comparable across classes.

Limitations of Existing Approaches and Thesis Motivation. Despite their empirical strength, current approaches suffer from several key limitations:

1. **Lack of Rigor in Metric Frameworks**: The covariance normalisation used by FeCAM is heuristic, and the Gaussian log-normalisation factor—the log-determinant term required by a proper multivariate Gaussian log-likelihood—is entirely omitted from its scoring function, preventing it from being interpreted as a fully probabilistic classifier.
2. **Loss of Generative Interpretability**: RanPAC's random projection introduces additional hyperparameters and breaks the direct correspondence between accumulated class statistics and an explicit probability density, since the projected space no longer admits a simple generative interpretation.
3. **Unverified Assumptions**: Broadly, none of the methods reviewed in this section rigorously validates whether the Gaussian assumption is statistically justified in foundation-model embedding spaces, nor do they provide a direct theoretical link between inter-class separability and achievable accuracy.

Chapter 3

Modelling Class Densities in a Frozen Feature Space

3.1 Proposed Architecture

As shown in Figure 3.1, our CIL framework is built upon a simple yet structurally motivated decomposition: (i) a frozen feature extractor based on a visual foundation model, and (ii) a probabilistic classification layer that models each object category through an independent Gaussian distribution in the embedding space. Given an input image, the feature extractor maps it to a fixed high-dimensional representation. For each class, we maintain a class-conditional multivariate Gaussian characterized by its mean vector and covariance matrix, estimated incrementally from observed samples. Classification is performed according to the maximum likelihood principle, assigning the input to the class whose Gaussian model yields the highest likelihood under the stationary embedding representation. This modular separation between representation learning and statistical modeling enables incremental updates without backpropagation or parameter adaptation: adding a new class requires only estimating its distribution in the fixed feature space, while previously learned classes remain unaffected at the representation level.

3.1.1 Likelihood-Based Classification

Given an input image $x \in \mathcal{X}$, the frozen foundation model $\mathcal{F}$ produces a fixed-dimensional embedding

$$\vec{x} = \mathcal{F}(x) \in \mathbb{R}^d \tag{3.1}$$

and all learning is performed directly in the stationary feature space $\mathbb{R}^d$. We adopt a generative modelling framework and assume class-conditional Gaussian feature distributions:

$$p(\vec{x} \mid y = c) = \mathcal{N}(\vec{x} \mid \boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c), \tag{3.2}$$

where $\boldsymbol{\mu}_c \in \mathbb{R}^d$ and $\boldsymbol{\Sigma}_c \succ 0$ are the class-specific mean and covariance matrix, estimated incrementally from observed samples. Classification is performed via Maximum A Posteriori

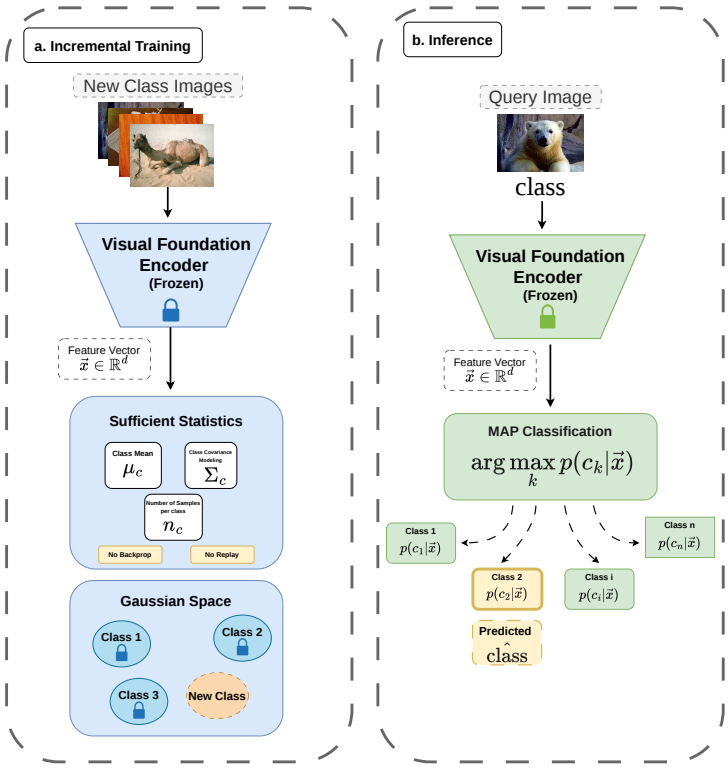


Figure 3.1: Overview of the proposed framework, reproduced from the peer-reviewed article that accompanies this thesis. A frozen foundation encoder maps each input image to a fixed embedding $\vec{x} \in \mathbb{R}^d$. Each class is represented by a Gaussian density (mean and covariance) that is updated incrementally without backpropagation. At inference, classification follows the MAP rule under the learned class-conditional densities. This diagram supersedes the general conceptual schematic used purely for narrative context in the Introduction (Figure 1.1), adding explicit detail on the recursive statistic-update mechanism shared by all covariance parameterisations introduced in Section 3.2.

(MAP) inference:

$$\hat{y} = \arg\max_{c \in \mathcal{Y}} \left[\log p(\vec{x} \mid y = c) + \log \pi_c \right], \tag{3.3}$$

where $\pi_c = p(y = c)$ is the class prior. Under a uniform prior and the Gaussian density model, the decision rule is equivalent to the quadratic discriminant:

$$\hat{y} = \arg\min_{c \in \mathcal{Y}} \left[(\vec{x} - \boldsymbol{\mu}_c)^T \boldsymbol{\Sigma}_c^{-1} (\vec{x} - \boldsymbol{\mu}_c) + \log |\boldsymbol{\Sigma}_c| \right]. \tag{3.4}$$

This quadratic decision function is the core of the proposed framework. The Mahalanobis term $(\vec{x} - \boldsymbol{\mu}_c)^T \boldsymbol{\Sigma}_c^{-1} (\vec{x} - \boldsymbol{\mu}_c)$ encodes the *shape* of the class distribution while $\log |\boldsymbol{\Sigma}_c|$ encodes its *volume*. The covariance parameterisations introduced in Section 3.2 are specialisations of Equation (3.4) that differ in the structure imposed on $\boldsymbol{\Sigma}_c$; the interplay between the shape and volume components is the central design axis of the covariance estimation study and is examined in detail in the analysis of results.

Adding a new class to the system requires only estimating the sufficient statistics $(\boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c)$ in the fixed feature space; previously learned class statistics are never modified. This for-

mulation therefore enables genuinely incremental updates without backpropagation, replay buffers, or exemplar storage.

3.1.2 Datasets and Experimental Protocol

We follow the standard Continual Learning (CL) protocol where the model learns a sequence of disjoint tasks and calculates the average accuracy and the forgetting. The specific configurations for the datasets are as follows:

- **Split CIFAR-100:** This dataset is a widely utilized benchmark in the continual learning literature. It partitions the original CIFAR-100 [27] dataset into 10 disjoint tasks, with each task containing 10 distinct classes.
- **Split ImageNet-R:** This benchmark is constructed from ImageNet-R [26], a dataset containing renditions of ImageNet classes such as art, cartoons, and sketches. We randomly divide the 200 classes into 10 sequential tasks, with 20 classes per task, introducing a significant domain shift challenge.
- **Split Caltech-256:** Designed to evaluate the model on higher-resolution images with greater object variability, this benchmark utilises the Caltech-256 [28] dataset. The first 250 of the original 256 categories are used and sequentially divided into 10 disjoint tasks of 25 classes each; the remaining six categories are discarded so that the benchmark has a fixed, balanced number of classes per task. The resulting protocol contains approximately 28 100 images with an average of $\overline{n_c} \approx 112$ training samples per class, which makes the per-class budget the smallest of the three constituent datasets and stresses the high-dimensional covariance estimators introduced in Section 3.2.
- **Split Joint Dataset:** This benchmark serves as a comprehensive stress test by aggregating the 30 tasks from the three aforementioned datasets into a single, heterogeneous stream. The tasks are presented to the system in a randomized order, forcing the model to navigate extreme distribution shifts between natural, artistic, and high-resolution images. This long-horizon sequence is specifically designed to evaluate the model's resilience against catastrophic forgetting over an extended period of diverse learning objectives.

3.1.3 Model Metrics

As mentioned, to quantify performance, SOTA works [44] utilize two standard metrics derived from the test accuracy scores: the average accuracy and the average forgetting.

On the one hand, **Average Accuracy (A_t)**, presented on equation (3.5), measures overall performance after task t. It is the mean test accuracy across all tasks learned up to that point.

$$A_t = \frac{1}{t} \sum_{\tau=1}^{t} S_{t,\tau} \tag{3.5}$$

On the other hand, **Forgetting (F_t)**, presented on equation (3.6), quantifies how much the model has forgotten earlier tasks. It computes, for each past task, how much accuracy has dropped compared to its best historical score, and averages those drops.

$$F_t = \frac{1}{t-1} \sum_{\tau=1}^{t-1} \left(\max_{k \leq t-1} S_{k,\tau} - S_{t,\tau} \right) \tag{3.6}$$

Regarding forgetting, the proposed framework does not exhibit forgetting in the conventional sense, as previously learned class statistics are never modified when new classes are introduced. The measured forgetting values therefore primarily reflect the increased classification difficulty as the number of competing classes grows, rather than degradation of previously learned representations.

With the architecture, benchmarks and evaluation metrics now in place, the remainder of this chapter follows a single, consistent narrative arc that Chapter 4 mirrors section by section: it opens with the covariance-estimation methodology and its mixture-of-Gaussians extensions, then introduces the probabilistic confidence and calibration metrics that complement raw accuracy, followed by the incremental class-inclusion protocol that stresses the growing decision surface. It then turns to the inter-class pairwise Bhattacharyya distance, which quantifies the separability of the learned class-conditional Gaussians, before examining the sensitivity of the framework to the choice of upstream backbone and the spectral structure of the embedding space it produces. The chapter closes with the statistical validation of the Gaussianity assumption on which the whole framework rests.

3.2 Covariance Estimation

Due to the high dimensionality of the extracted feature ($\vec{x} \in \mathbb{R}^d$), the empirical sample covariance matrix (SCM) is a poor estimator for each class: its eigenvalues are biased, and the matrix is singular whenever $n_c < d$. In our setting $d = 768$ while n_c ranges from approximately 40 to 400 depending on the benchmark, so the matrix always has at most $n_c - 1$ non-zero eigenvalues out of $d = 768$, leaving $d - (n_c - 1)$ eigenvalues that are structurally zero before any regularisation is applied. The Gaussian modelling assumption that motivates the covariance-based classifiers introduced below is formally validated later in Section 3.10; here we focus on how to obtain a well-conditioned covariance estimate under that assumption. Our benchmarks are tested using different covariance estimation methods:

3.2.1 Diagonal Covariances

In this configuration, we assume that the features within each class c are independent, leading to a diagonal covariance matrix $\mathbf{\Sigma}_c = \text{diag}(\sigma_{c,1}^2, \sigma_{c,2}^2, \ldots, \sigma_{c,d}^2)$. This simplification significantly reduces the number of parameters to estimate from $O(d^2)$ to $O(d)$, which is particularly advantageous when the dimensionality $d = 768$ is large relative to the number of samples per class. For each class $c \in \{1, \ldots, C\}$, the mean vector $\boldsymbol{\mu}_c$ and the diagonal elements of $\mathbf{\Sigma}_c$ are computed as:

$$\boldsymbol{\mu}_c = \frac{1}{n_c} \sum_{i=1}^{n_c} \vec{x}_i, \quad \sigma_{c,j}^2 = \frac{1}{n_c - 1} \sum_{i=1}^{n_c} (x_{i,j} - \mu_{c,j})^2 + \varepsilon \tag{3.7}$$

where $\varepsilon = 10^{-6}$ is a small constant added to ensure numerical stability and strictly positive variances. The log-likelihood of a test sample $\vec{x}$ belonging to class c is determined by the discriminant function:

$$g_c(\vec{x}) = \ln P(\vec{x}|c) + \ln P(c) \tag{3.8}$$

Under the assumption of a uniform prior $P(c)$ and Gaussian distribution, this simplifies to:

$$g_c(\vec{x}) = -\frac{1}{2} \sum_{j=1}^{d} \left(\ln(2\pi\sigma_{c,j}^2) + \frac{(x_j - \mu_{c,j})^2}{\sigma_{c,j}^2} \right) \tag{3.9}$$

3.2.2 Full Covariances

To account for the dependencies between different dimensions of the feature vector, we estimate a full covariance matrix $\mathbf{\Sigma}_c \in \mathbb{R}^{d \times d}$ for each class c. While this provides a richer representation of the class distribution, the estimation of d^2 parameters is prone to ill-conditioning when the number of samples n_c is limited.The mean vector $\boldsymbol{\mu}_c$ and the empirical covariance $\mathbf{\Sigma}_c$ are calculated as:

$$\boldsymbol{\mu}_c = \frac{1}{n_c} \sum_{i=1}^{n_c} \vec{x}_i, \quad \mathbf{\Sigma}_c = \frac{1}{n_c - 1} \sum_{i=1}^{n_c} (\vec{x}_i - \boldsymbol{\mu}_c)(\vec{x}_i - \boldsymbol{\mu}_c)^T \tag{3.10}$$

To ensure that the matrix is invertible and to improve numerical stability, a regularization is applied by adding a small identity term to the diagonal: $\hat{\mathbf{\Sigma}}_c = \mathbf{\Sigma}_c + \varepsilon I$, with $\varepsilon = 10^{-6}$.

For obtaining the precision matrix Q_c (the inverse of the covariance matrix $\hat{\mathbf{\Sigma}}_c$), an approach based on Cholesky decompositions is used. The covariance $\hat{\mathbf{\Sigma}}_c$ is decomposed as

$$\hat{\mathbf{\Sigma}}_c = C_c C_c^T, \qquad C_c L_c = I, \tag{3.11}$$

with C_c lower triangular and I the identity, so that $L_c = C_c^{-1}$ is the Cholesky factor of the precision matrix $Q_c = L_c^T L_c$.

Then, the squared Mahalanobis distance and the log-determinant are computed from $\boldsymbol{\mu}_c$ and L_c as:

$$d_M(\vec{x}, \boldsymbol{\mu}_c, L_c) = (\vec{x}L_c - \boldsymbol{\mu}_c L_c)^T (\vec{x}L_c - \boldsymbol{\mu}_c L_c) \tag{3.12}$$

$$\ln |\hat{\boldsymbol{\Sigma}}_c| = 2 \sum_{i=1}^{d} \ln(L_{c\,i,i}) \tag{3.13}$$

3.2.3 Hybrid Covariance

The hybrid covariance estimator decouples the squared Mahalanobis distance from its associated normalisation constant. For each class c, the full covariance matrix $\hat{\boldsymbol{\Sigma}}_c$ is estimated and regularised exactly as in Section 3.2.2, and the squared Mahalanobis distance $d_M(\vec{x}, \mu_c, L_c)$ is computed using the same Cholesky-based procedure of Equations (3.11)–(3.12), so that the *shape* of the discriminant retains the full off-diagonal structure of the empirical covariance. However, the log-determinant term—which represents the *volume* of the distribution—is computed using only the diagonal elements (variances) of the covariance matrix:

$$\ln |\hat{\boldsymbol{\Sigma}}_c|_{\text{diag}} = \sum_{j=1}^{d} \ln(\sigma_{c,j}^2), \tag{3.14}$$

where $\sigma_{c,j}^2$ denotes the j-th diagonal entry of $\hat{\boldsymbol{\Sigma}}_c$. The resulting discriminant function is defined as:

$$g_c(\vec{x}) = -\frac{1}{2} \left(d \ln(2\pi) + \sum_{j=1}^{d} \ln(\sigma_{c,j}^2) + d_M(\vec{x}, \boldsymbol{\mu}_c, L_c) \right). \tag{3.15}$$

This formulation avoids the numerical issues that arise when near-zero eigenvalues of $\hat{\boldsymbol{\Sigma}}_c$ artificially inflate or destabilise the exact log-determinant, at the cost of using a diagonal approximation for the volume term only.

3.2.4 OAS Covariance

Oracle Approximating Shrinkage (OAS) is a shrinkage technique that computes an structurally optimized covariance matrix by finding an optimal compromise between the sample covariance matrix and a structured target (the identity matrix scaled by the average trace).The OAS estimator provides a more accurate estimation of the covariance $\boldsymbol{\Sigma}_c$ by minimizing the Mean Squared Error (MSE) under the assumption of Gaussianity. The estimated covariance is defined as:

$$\hat{\mathbf{\Sigma}}_{\text{OAS},c} = (1 - \rho_c)\hat{\mathbf{\Sigma}}_{\text{SCM},c} + \rho_c \hat{F}_c \tag{3.16}$$

where ρ_c is the shrinkage coefficient calculated specifically for class c, $\hat{\mathbf{\Sigma}}_{\text{SCM},c}$ is the sample covariance matrix, and $\hat{F}_c = \frac{\text{tr}(\hat{\mathbf{\Sigma}}_{\text{SCM},c})}{d} I$ is the shrinkage target.In our implementation, we compute the precision matrix $\Psi_c = \hat{\mathbf{\Sigma}}_{\text{OAS},c}^{-1}$ directly via the OAS algorithm. This precision matrix is then used to calculate the Mahalanobis distance in the log-likelihood discriminant function:

$$g_c(\vec{x}) = \ln A_c - \frac{1}{2}(\vec{x} - \boldsymbol{\mu}_c)^T \Psi_c (\vec{x} - \boldsymbol{\mu}_c) \tag{3.17}$$

where $\ln A_c$ is the normalization constant derived from the log-determinant of the OAS-estimated covariance matrix, evaluated numerically via the stable log-determinant of the eigendecomposition. This approach ensures that the model captures class-specific correlations while remaining robust to the "curse of dimensionality."

3.2.5 Ledoit-Wolff

The Ledoit-Wolf Diagonal Anchor estimator addresses the instability of the sample covariance matrix S by shrinking it toward a target matrix T.

The selection of a target matrix T represents a prior assumption regarding the feature distribution. Both methods stabilize the estimator by pulling the noisy off-diagonal elements of the sample covariance matrix S toward zero, but they differ in how they treat the diagonal elements (the variances).

Diagonal Target

The diagonal anchor is defined as $T_{\text{diag}} = \text{diag}(s_{11}, s_{22}, \ldots, s_{dd})$. This approach assumes that the features are independent but allows for heteroscedasticity, meaning each dimension can have a different variance. Shrinkage toward this target preserves the scale of each individual feature as observed in the sample data while suppressing only the correlations. This is particularly useful when certain dimensions of the DINOv2 embeddings are naturally more informative or volatile than others.

Identity Target

The identity anchor is more restrictive, defined as $T_{\text{ident}} = \mu I$, where $\mu = \frac{1}{d}\text{tr}(S)$ is the average variance across all dimensions. This target assumes that the features are not only independent but also share a common variance (homoscedasticity). The resulting distribution is spherical. Because the embeddings used in this work are L2-normalized, the variances across dimensions tend to be similar, making the identity anchor a highly robust regularizer that prevents any single dimension from disproportionately influencing the distance metric.

Structural Summary

The fundamental difference between these two estimators lies in the flexibility of the variance components:

- **Diagonal Anchor:** $\hat{\Sigma}_{ij} = (1-\gamma)s_{ij}$ for $i \neq j$, and $\hat{\Sigma}_{ii} = s_{ii}$. It yields an axis-aligned ellipsoidal distribution.
- **Identity Anchor:** $\hat{\Sigma}_{ij} = (1-\gamma)s_{ij}$ for $i \neq j$, and $\hat{\Sigma}_{ii} = (1-\gamma)s_{ii} + \gamma\mu$. It yields a distribution that is pulled toward a perfectly spherical geometry.

3.2.6 Shared Covariance

The Shared Covariance framework assumes that the internal noise or feature variability is homoscedastic across all classes. In this paradigm, all classes share a single covariance matrix $\mathbf{\Sigma}$, which captures the global structure of the feature space.

A key aspect of this framework in the continual learning setting is that $\mathbf{\Sigma}$ is re-estimated at the end of each task t using only the classes observed so far, $\mathcal{C}_t = \{1, \ldots, C_t\}$.

3.3 Mixture of Gaussians

In order to capture potential multi-modality within the feature space of a single class, we employ Gaussian Mixture Models (GMM). We implemented the following models using the previously defined diagonal, full and hybrid covariances approaches.

3.3.1 Mixture of 2 Gaussians

Forcing each class to be defined as a GMM with $K = 2$ components, each class c can be modeled by a probability density function:

$$p(\vec{x}|c) = \sum_{k=1}^{K} \pi_{c,k} \mathcal{N}(\vec{x}|\boldsymbol{\mu}_{c,k}, \mathbf{\Sigma}_{c,k}) \tag{3.18}$$

where $\pi_{c,k}$ represents the mixing weight of the k-th component for class c, subject to $\sum \pi_{c,k} = 1$. In this specific variant, the covariance matrices $\mathbf{\Sigma}_{c,k}$ are restricted to be diagonal to maintain computational efficiency and prevent over-fitting in the high-dimensional DINOv2 space.The parameters $\{\pi_{c,k}, \boldsymbol{\mu}_{c,k}, \mathbf{\Sigma}_{c,k}\}$ are estimated using the Expectation-Maximization (EM) algorithm. To ensure numerical stability, a small constant $\varepsilon = 10^{-5}$ is added to the diagonal of each covariance matrix. The log-likelihood for a test sample $\vec{x}$ is calculated using the *LogSumExp* operator to avoid underflow:

$$\ln p(\vec{x}|c) = \text{lnaddexp}\left(\ln(\pi_{c,1}\mathcal{N}(\vec{x}|\boldsymbol{\mu}_{c,1}, \boldsymbol{\Sigma}_{c,1})), \ln(\pi_{c,2}\mathcal{N}(\vec{x}|\boldsymbol{\mu}_{c,2}, \boldsymbol{\Sigma}_{c,2}))\right) \tag{3.19}$$

In the implementation, this is expanded as:

$$\ln p(\vec{x}|c) = \ln\left(\exp(\ln \pi_{c,1} + \ln A_{c,1} - \frac{1}{2}d_{M,1}) + \exp(\ln \pi_{c,2} + \ln A_{c,2} - \frac{1}{2}d_{M,2})\right) \tag{3.20}$$

where $d_{M,k}$ is the diagonal Mahalanobis distance to the k-th component and $\ln A_{c,k}$ is the normalization constant. This approach allows the model to better represent classes that may contain significantly different visual sub-types or clusters within the latent embedding space.

3.3.2 Mixture of up-to-4 Gaussians

To adapt to the varying complexity of different classes within the different datasets, we implement a flexible Gaussian Mixture Model where the number of components K_c for each class c is not fixed. Instead, the model selects the most appropriate number of components $K_c \in \{1, 2, 3, 4\}$ by minimizing the Bayesian Information Criterion (BIC).The BIC is defined as:

$$\text{BIC} = \ln(n)k - 2\ln(\hat{L}) \tag{3.21}$$

where n is the number of samples in the class, k is the number of free parameters to be estimated, and $\hat{L}$ is the maximized value of the likelihood function. By using diagonal covariance matrices, the number of parameters per component is kept low $(2d + 1)$, allowing the selection process to favor multi-modal distributions only when the increase in likelihood justifies the additional parameters. For each component k in class c, the diagonal covariance $\boldsymbol{\Sigma}_{c,k} = \text{diag}(\sigma^2_{c,k,1}, \ldots, \sigma^2_{c,k,d})$ is regularized with $\varepsilon = 10^{-6}$. The classification of a test sample $\vec{x}$ involves calculating the weighted log-likelihood across all active classes:

$$\ln p(\vec{x}|c) = \text{LSE}_{k=1}^{K_c}\left(\ln \pi_{c,k} - \frac{1}{2}\left(\sum_{j=1}^{d}\ln(2\pi\sigma^2_{c,k,j}) + \sum_{j=1}^{d}\frac{(x_j - \mu_{c,k,j})^2}{\sigma^2_{c,k,j}}\right)\right) \tag{3.22}$$

where LSE denotes the LogSumExp operator. This methodology allows for a more granular representation of class distributions—using a single Gaussian for simple, compact classes and up to four Gaussians for classes with high intra-class variance or distinct visual sub-clusters—while preserving the numerical efficiency of diagonal estimators.

3.4 Confidence and Calibration Metrics

While accuracy and forgetting measure the correctness of predictions, they do not account for the model's reliability or its tendency to over-confide in incorrect decisions—a common issue in Class-Incremental Learning. To address this, we introduce five complementary metrics that jointly characterise the probabilistic behaviour of the classifier: the Expected Calibration Error (ECE) measures the global alignment between confidence and accuracy; the Confidence on Errors (CoE) quantifies how dangerously overconfident the model is when it fails; Total Confidence (TC) and Probability Margin (PM) both capture the decisiveness of the posterior distribution from two complementary angles; and the Average Predictive Entropy (H) measures the spread of probability mass across all classes.

3.4.1 Expected Calibration Error (ECE)

This metric quantifies the mismatch between predictive confidence and empirical accuracy. We partition the predictions into M equally spaced bins based on their confidence scores. The ECE is defined as the weighted average of the absolute difference between the accuracy and confidence of each bin:

$$ECE_t = \sum_{m=1}^{M} \frac{|B_m|}{N} \left| acc(B_m) - conf(B_m) \right| \tag{3.23}$$

where N is the total number of samples, B_m is the set of indices of samples whose prediction confidence falls into the m-th interval, and $acc(B_m)$ and $conf(B_m)$ represent the mean accuracy and mean confidence within that bin, respectively.

3.4.2 Confidence on Errors (CoE)

To specifically analyze the model's behavior when it fails, we measure the average confidence assigned to the winning (but incorrect) class. High CoE indicates a model that is "dangerously wrong", which is a critical factor in safety-oriented applications. It is computed for a task t as:

$$CoE_t = \frac{1}{|\mathcal{E}_t|} \sum_{i \in \mathcal{E}_t} \max(\hat{p}_i) \tag{3.24}$$

where $\mathcal{E}_t$ is the set of misclassified samples and $\hat{p}_i$ is the predicted probability distribution for sample i.

3.4.3 Total Confidence (TC)

Total Confidence measures how certain the model is on average about its top-1 prediction, regardless of whether that prediction is correct:

$$TC_t = \frac{1}{N} \sum_{i=1}^{N} \max_c p_{i,c} \tag{3.25}$$

A value of $TC = 1$ means the model assigns all probability mass to a single class for every sample—it is always maximally decisive. A value close to $1/C$ would indicate near-uniform posteriors. TC is a measure of *decisiveness*, not correctness: a well-calibrated model should have $TC \approx A_T$ (confident when right, uncertain when wrong), whereas a pathologically overconfident model can reach $TC \approx 1$ even at low accuracy.

3.4.4 Probability Margin (PM)

The Probability Margin is the natural, non-logarithmic measure of how decisively the model separates its top-1 prediction from its second-best alternative:

$$PM_t = \frac{1}{N} \sum_{i=1}^{N} \Big(p_{i,(1)} - p_{i,(2)} \Big) \tag{3.26}$$

where $p_{i,(1)} \geq p_{i,(2)}$ are the two highest posterior probabilities for sample i. A margin near 1 means the model almost entirely ignores its second candidate; a margin near 0 means the model is nearly tied between its top-two guesses and one should treat the prediction with caution. PM is bounded in $[0, 1]$ and directly interpretable: a value of 0.7 means the winning class is, on average, assigned 70% more probability than the runner-up.

3.4.5 Average Predictive Entropy (H)

We utilize Shannon entropy to measure the uncertainty of the model's predictions. A higher entropy suggests that the model is distributing its probability mass across multiple classes, indicating lower certainty. For task t, the average entropy is calculated as:

$$H_t = \frac{1}{N} \sum_{i=1}^{N} \left(- \sum_{c=1}^{C} p_{i,c} \log(p_{i,c}) \right) \tag{3.27}$$

where $p_{i,c}$ is the probability assigned to class c for sample i.

Summary of calibration and uncertainty metrics

Table 3.1 collects all six metrics (accuracy and the five calibration/uncertainty indicators), their range, the direction of improvement, and the key question each one answers. Together they provide a multi-dimensional view of the model's reliability throughout continual learning: accuracy tells us *how often* the model is right; ECE tells us *how honest* its probabilities are; CoE tells us *how dangerous* its mistakes are; TC tells us *how decisive* it is overall; PM tells us *how clearly* it separates its top-two candidates; and entropy tells us *how spread* its uncertainty is across all classes.

Table 3.1: Summary of the probabilistic calibration and uncertainty metrics used in this work. "Better" indicates the direction of the preferred value; "context" means both high and low values are informative depending on whether predictions are correct.

Metric	**Symbol**	**Range**	**Better**	**Key question**
Accuracy	A_T	$[0, 1]$	$\uparrow$	What fraction of predictions are correct?
Expected Calibration Error	ECE	$[0, 1]$	$\downarrow$	How well does confidence match actual accuracy?
Confidence on Errors	CoE	$[0, 1]$	$\downarrow$	How confident is the model when it is *wrong*?
Total Confidence	TC	$[0, 1]$	context	How decisive is the model on its top-1 choice?
Probability Margin	PM	$[0, 1]$	context	How much more likely is top-1 than top-2?
Predictive Entropy	H	$[0, \log C]$	$\downarrow$	How spread is probability mass over all C classes?

3.5 Incremental Class Inclusion

Having defined the full family of covariance estimators in Section 3.2, we design a class-incremental evaluation protocol to systematically assess how each closed-form estimator scales as the overall classification task grows in complexity. Rather than measuring conventional catastrophic forgetting—given that each class-conditional Gaussian is estimated independently and frozen upon initialization—this incremental class inclusion experiment isolates the impact of a progressively hardening decision surface on classification performance. By sequentially expanding the set of active classes and tracking the evolution of average accuracy over the growing pool of target categories, we can observe how the physical expansion of the label space translates into geometric crowding within the underlying feature space. This experimental design directly bridges empirical performance degradation with the architectural geometry of the learned representations, paving the way to quantify how the introduction of new class-conditional distributions compresses the margins between prototype boundaries and intensifies pairwise class confusion.

3.6 Inter-Class Structural Separability Analysis

To understand how the Bayesian decision surface scales and characterize the geometric crowding of the embedding space as new tasks are incorporated, we evaluate inter-class relationships using two complementary metrics: the probabilistic Bhattacharyya distance and the geometric Euclidean distance.

3.6.1 Pairwise Bhattacharyya Distance

The *inter-class* pairwise Bhattacharyya distance measures how distinguishable any two class-conditional Gaussians are from one another. The Bhattacharyya distance is symmetric and provides a direct connection to the Bayes error rate via the Bhattacharyya coefficient $\rho_{ij} = e^{-D_B(i,j)}$. Under the diagonal-Gaussian parameterisation, the Bhattacharyya distance between classes i and j is defined as:

$$D_B(i,j) = \frac{1}{8}(\boldsymbol{\mu}_i - \boldsymbol{\mu}_j)^\top \bar{\Sigma}^{-1}(\boldsymbol{\mu}_i - \boldsymbol{\mu}_j) + \frac{1}{2}\ln\left(\frac{\det \bar{\Sigma}}{\sqrt{\det \Sigma_i \, \det \Sigma_j}}\right), \tag{3.28}$$

where $\bar{\Sigma} = (\Sigma_i + \Sigma_j)/2$. Because both Σ_i and $\bar{\Sigma}$ are diagonal, all matrix operations reduce to element-wise arithmetic: determinants become products of the diagonal entries (or sums of logs), and the matrix inverse reduces to element-wise reciprocals. A value of $D_B = 0$ indicates complete distributional overlap; larger values correspond to greater separability and a lower Bhattacharyya coefficient $\rho = e^{-D_B}$.

3.6.2 Pairwise Mean Euclidean Distance

While the Bhattacharyya distance jointly accounts for shifts in both first-order (means) and second-order (covariances) statistics, it can mask whether class texturing or raw prototype drift is the dominant factor in class confusion. Therefore, we isolate first-order geometric separation by calculating the plain pairwise Euclidean distance between the estimated class mean vectors $\boldsymbol{\mu}_i$ and $\boldsymbol{\mu}_j$:

$$d_E(i,j) = \|\boldsymbol{\mu}_i - \boldsymbol{\mu}_j\|_2 = \sqrt{\sum_{k=1}^{d}(\mu_{i,k} - \mu_{j,k})^2}, \tag{3.29}$$

where d is the nominal dimensionality of the visual foundation embedding space.

3.6.3 Stream Statistics and Trajectory Tracking

As the total number of observed classes k scales throughout the continual learning stream, we track three distinct statistics for both D_B and d_E across the expanding class-pair pool:

(a) The **macro-averaged pairwise distance** ($\overline{D_B}$ and $\overline{d_E}$) flanked by a $[\mathrm{P}_{10}, \mathrm{P}_{90}]$ percentile band to observe global drift.

(b) The **minimum pairwise distance** ($\min D_B$ and $\min d_E$), which explicitly identifies the hardest, most confusable class pair in the system.

(c) The **cumulative count of overlapping pairs** falling below critical proximity thresholds (e.g., $D_B \in \{10, 50, 100\}$ and $d_E \in \{0.3, 0.5, 0.7\}$) to capture the heavy-tailed growth of localized confusion sinks.

Together, these metrics bridge empirical classification failure modes with the underlying statistical architecture of the learned representations.

3.7 Backbones Comparison

Throughout this work, DINOv2 [11] is used as the default backbone feature extractor for the Gaussian-based classifiers because of its self-supervised training regime and the strong empirical evidence of approximately Gaussian, well-conditioned class-conditional distributions presented in Section 3.10. However, the proposed framework is agnostic to the specific feature extractor used: any frozen encoder that produces fixed-dimensional embeddings can plug in unchanged into the downstream Gaussian classifier. To quantify the dependence of the framework on the choice of backbone, we further analyse the impact of the backbone architecture by re-running the Diag-Cov and Full-Cov variants using two supervised Vision Transformers from the `timm` library [56]: **ViT-base/16** (ViT-b, $d = 768$) and **ViT-medium/16** (ViT-m, $d = 512$), both pre-trained on ImageNet-21k [25]. The covariance estimation, hyper-parameters and evaluation protocol are kept identical across backbones to isolate the effect of the upstream representation; the results are reported in Section 4.6.

This concludes the comparison of the framework's sensitivity to the choice of upstream backbone. The next section shifts focus from the classifier back to the representation itself, characterising the global spectral structure of the embedding space that underlies every covariance estimator considered above.

3.8 Spectral Geometry and Dimension-Reduction Probing

Having isolated the visual performance differentials across upstream backbones, we shift our focus from downstream classification mechanics to the intrinsic geometric structure of the embedding space itself. This structural investigation is divided into a descriptive spectral characterization and a downstream low-dimensional classification stress test.

3.9 Spectral Geometry and Dimension-Reduction Probing

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3.9.1 Spectral Analysis of the Foundation Embeddings

To characterize the global variance distribution of the foundation feature space independently of individual class-conditional labels, we compute the empirical eigenspectrum of the ℓ_2-normalised embeddings pooled across the entirety of each benchmark's training split. The $d = 768$ eigenvalues ($\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0$) of the pooled data covariance matrix are sorted in descending order to analyze individual percentage-of-variance contributions and the cumulative explained variance.

This diagnostic explicitly measures the *effective intrinsic dimensionality* of the shared representation. A rapid, steep eigenvalue decay implies that the visual discriminative signal is concentrated within a tight, low-dimensional active subspace, pointing to high structural redundancy in the ambient space. Conversely, a slow, uniform decay indicates an isotropic distribution where discriminative information is scattered widely across the full ambient space. Furthermore, this global spectral profile establishes a strict theoretical upper bound on the effective rank available to the localized, per-class covariance estimators, as a class-specific model can only populate directional parameters already spanned by the pooled population spectrum.

3.9.2 Downstream PCA Dimension-Reduction Probing

While the spectral analysis in Section 3.9.1 describes the native distribution of the features, it does not evaluate whether a low-dimensional concentration can be systematically exploited to alleviate the high-dimensional small-sample rank deficiency ($n_c < d$) highlighted in Section 3.2. To establish a diagnostic boundary on performance under forced compression, we introduce a downstream PCA dimension-reduction test.

Rather than fitting a localized, data-dependent Principal Component Analysis on each target stream—which would violate the strict class-incremental constraints by altering the projection matrix over time—we utilize a fixed, out-of-distribution linear projection matrix $\mathbf{P}_k \in \mathbb{R}^{d \times k}$. This projection matrix is pre-trained exclusively on the supervised distribution of the *ImageNet-21k* corpus, acting as a general-purpose, dataset-agnostic structural probe.

For a chosen reduced target dimensionality $k \in \{64, 256\}$, the ℓ_2-normalized foundation features $\mathbf{x} \in \mathbb{R}^d$ are projected into a compressed subspace via:

$$\tilde{\mathbf{x}} = \mathbf{P}_k^\top (\mathbf{x} - \boldsymbol{\mu}_{\text{global}}), \tag{3.30}$$

where $\boldsymbol{\mu}_{\text{global}}$ is the stationary global mean of the pre-training corpus. The class-conditional Gaussian classifiers (both Diagonal and Full covariance parameterisations) are subsequently initialized, trained, and evaluated entirely within this frozen, low-dimensional coordinate system.

The relative scaling of accuracy and estimator stability across these reduced dimensions serves as a multi-faceted probe into the underlying geometry:

The robustness of downstream CIL performance under forced compression ($k \ll d$) demonstrates an alignment between the current discriminative signal and the leading prin-

cipal axes of the supervised out-of-distribution probe, confirming that trailing variances act primarily as non-essential noise. Conversely, a pronounced accuracy degradation alongside decreasing dimensionality reveals that fine-grained inter-class decision boundaries remain heavily reliant on the long tail of the spectrum, rendering the full ambient high-dimensional space mandatory for preserving the class representations.

Furthermore, the relative performance gap between the Full and Diagonal covariance models exposes the interaction between subspace compression and data scarcity. A narrowing of the gap at lower dimensionalities indicates that the regularizing effect of the projection successfully mitigates the small-sample rank deficiency of the SCM. In contrast, a wider divergence or a systemic drop in Full covariance precision points to a structural distortion of underlying class-conditional correlations induced by the linear transformation.

3.10 Analysis of Gaussianity Assumptions

The covariance estimators and downstream classification frameworks rest on the working assumption that class-conditional features are approximately Gaussian in the embedding space of visual foundation models (such as DINOv2). To validate this assumption empirically without relying on manual semantic labels, we employ a multi-layered statistical evaluation on the training split of our datasets.

Our validation is structured across two levels:

1. **Per-Dimension (Univariate) Gaussianity:** Evaluated using moment-based shape diagnostics, the formal D'Agostino–Pearson omnibus normality test, and a coordinate-wise application of the Henze–Zirkler (HZ) test.
2. **Per-Class (Multivariate) Gaussianity:** Evaluated in the native high-dimensional feature space ($d = 768$) using a specialized diagonal-covariance adaptation of the HZ test to circumvent sample-size rank deficiencies.

3.10.1 Per-Dimension Gaussianity: Moments and Formal Tests

To verify Gaussianity at the level of individual feature coordinates, we apply three complementary indicators to every (class c, dimension j) pair.

Shape moments.

Following the normality diagnostics of Pearson [4] and D'Agostino and Pearson [12, 13], we compute the sample skewness and excess kurtosis for each class c along each of the d coordinates:

$$\widehat{\mathrm{Skew}}_{c,j} = \frac{\frac{1}{n_c}\sum_i (x_{i,j} - \mu_{c,j})^3}{\hat{\sigma}_{c,j}^3}, \qquad \widehat{\mathrm{Kurt}}_{c,j} = \frac{\frac{1}{n_c}\sum_i (x_{i,j} - \mu_{c,j})^4}{\hat{\sigma}_{c,j}^4} - 3, \tag{3.31}$$

where both quantities vanish exactly for a theoretical Gaussian distribution [4, 3].

D'Agostino–Pearson omnibus test.

The moments alone do not constitute a formal statistical test. We therefore apply the D'Agostino–Pearson omnibus test to every (class, dimension) pair [12, 13]. The test combines the two shape statistics into a single composite statistic:

$$K^2 \;=\; Z_1\Big(\widehat{\mathrm{Skew}}\Big)^2 + Z_2\Big(\widehat{\mathrm{Kurt}}\Big)^2 \;\sim\; \chi^2_2 \quad \text{under } H_0, \tag{3.32}$$

where Z_1 and Z_2 are normalised transformations of the skewness and kurtosis estimators respectively, and H_0 is the hypothesis that the marginal data are normally distributed. We reject H_0 at significance level $\alpha = 0.05$.

Relaxed criterion.

Because the omnibus test is highly sensitive to larger sample sizes n_c (where minor, non-critical departures from perfect normality still accumulate into significant K^2 values), we supplement it with a size-independent relaxed indicator. A dimension is classified as near-Gaussian if:

$$|\mathrm{Skew}| < 1 \quad \text{and} \quad |\mathrm{Kurt}| < 1 \tag{3.33}$$

hold simultaneously. This relaxed criterion acts alongside the strict omnibus test to establish lower and upper bounds on the proportion of approximately Gaussian coordinates.

3.10.2 Per-Class Gaussianity: Henze–Zirkler Test

Per-dimension normality is necessary but not sufficient for the joint d-dimensional distribution to be Gaussian, as coordinates can be marginally normal while departing from joint normality due to non-linear dependencies. We therefore complement the univariate analysis with the multivariate Henze–Zirkler (HZ) test [14].

Diagonal-covariance HZ at native resolution.

Computing a standard full-covariance HZ test is mathematically intractable in high-dimensional spaces where the per-class sample size is smaller than the feature dimensionality ($n_c < d$). Under these conditions, the empirical covariance matrix becomes singular and non-invertible, preventing the calculation of standard Mahalanobis distances.

To circumvent this rank deficiency without resorting to destructive dimensionality reduction, we formulate a diagonal-covariance adaptation of the HZ test. The required Mahalanobis distances are computed analytically by inverting only the d per-coordinate sample

variances (effectively regularizing the space), with a numerical floor of $\varepsilon = 10^{-10}$ applied to guard against inactive dimensions:

$$\mathbb{E}[T_{\mathrm{HZ}}] \;=\; 1 - (1+2b^2)^{-d/2} \;\to\; 1 \quad \text{as } d \to \infty, \tag{3.34}$$

meaning that values concentrated near $T_{\mathrm{HZ}} \approx 1$ serve as the Gaussian reference under the diagonal model.

3.10.3 Univariate Henze–Zirkler Test per Dimension

We bridge the univariate and multivariate analyses by introducing a univariate HZ analysis applied independently to every (class c, dimension j) pair. This provides a stricter hypothesis-based criterion than the D'Agostino–Pearson test, while remaining more interpretable than the full-dimensional multivariate result.

Bandwidth. For $d = 1$, the standard HZ smoothing parameter reduces to:

$$b^2 \;=\; \frac{1}{2}\left(\frac{4}{3}\right)^{2/5} n_c^{-2/5}. \tag{3.35}$$

Efficient vectorised computation. To handle all d dimensions simultaneously without forming prohibitive $n_c \times n_c$ distance matrices, the pairwise kernel term in T_{HZ} is expanded via its Taylor series around $b^2 x_j x_k = 0$:

$$A_m \;=\; \frac{1}{n_c^2}\sum_{j,k} e^{-\frac{b^2}{2}(z_{j,m}-z_{k,m})^2} \;=\; \frac{1}{n_c^2}\sum_{\ell=0}^{L} \frac{b^{2\ell}}{\ell!}\, S_\ell^2 \;+\; \mathcal{O}\big((b^2)^{L+1}\big)\,, \tag{3.36}$$

where $S_\ell = \sum_j v_{j,m}\, z_{j,m}^{\ell}$, $v_{j,m} = \exp(-b^2 z_{j,m}^2/2)$, and $L = 8$ terms are used to keep the approximation error below 10^{-6}.

Expected value under H_0 and p-value. For a general d-dimensional problem, the asymptotic expected value of T_{HZ} under exact normality is:

$$\mathbb{E}_{H_0}[T_{\mathrm{HZ}}] \;=\; 1 - (1+2b^2)^{-d/2}, \tag{3.37}$$

which is the reference used for the multivariate diagonal test of Section 3.10.2 ($d = 768$): even a modest bandwidth causes $(1+2b^2)^{-d/2} \to 0$ for large d, driving $\mathbb{E}_{H_0}[T_{\mathrm{HZ}}] \to 1$.

For the univariate case ($d = 1$) treated in this section, we instead use the exact finite-sample expectation, which includes higher-order correction terms that the large-d asymptotic approximation of Equation (3.37) discards:

$$\mathbb{E}_{H_0}[T_{\mathrm{HZ}}] \;=\; 1 - a^{-1/2}\left(1 + \frac{b^2}{a} + \frac{3b^4}{4a^2}\right), \qquad a = 1 + 2b^2, \tag{3.38}$$

with b^2 given by the bandwidth of Equation (3.35).

Under the null hypothesis H_0 of perfect univariate normality, we theoretically expect the empirical statistic T_{HZ} to be close to 0. For the class sizes encountered here ($n_c \in [60, 400]$), the adaptive bandwidth is small ($b^2 \approx 0.04$), which yields a baseline expectation of $\mathbb{E}_{H_0}[T_{\mathrm{HZ}}] \approx 0.04$. Consequently, any coordinate that is genuinely Gaussian will yield an empirical T_{HZ} value fluctuating tightly around this near-zero baseline, in contrast to the high-dimensional multivariate case where $T_{\mathrm{HZ}} \to 1$.

Chapter 4

Experimental Results and Analysis

This chapter presents the quantitative results of the experimental campaign together with their interpretation, topic by topic. Each section below first reports the relevant measurements and then, immediately afterwards, discusses the structural reasons behind the observed behaviour, so that evidence and analysis can be read side by side. The order of the sections mirrors the order in which the corresponding topics are introduced in the methodology (Chapter 3). Section 4.1 characterises the four benchmarks used throughout the thesis, including the empirical Gaussianity statistics that later sections validate formally. Section 4.2 reports continual-learning metrics for the covariance estimators and mixture-of-Gaussian variants against prompt-based and analytic state-of-the-art baselines, together with the volume–shape trade-off that explains their relative ordering, the numerical correction of the Full-Cov and Hybrid-Cov estimators and its empirical consequences, the effect of the log-normalisation constant, the score-component distributions, linear shrinkage, mixture models, and the framework's positioning against prompt-based baselines. Section 4.3 reports the probabilistic confidence and calibration metrics and interprets them as a further quality of the volume term. Section 4.4 examines how average accuracy evolves as the pool of active classes is incrementally expanded. Section 4.5 quantifies inter-class separability via the pairwise Bhattacharyya distance and interprets the resulting separability curves. Section 4.6 ablates the choice of feature extractor and discusses the framework's sensitivity to this choice. Section 4.7 characterises the PCA spectral decay of the DINOv2 embedding space and its consequence for PCA-reduced continual-learning accuracy. Section 4.8 reports the statistical validation of the Gaussianity assumption, combining moment-based diagnostics with the formal D'Agostino–Pearson and Henze–Zirkler hypothesis tests, and interprets what this validation implies for the modelling choices made throughout. Section 4.10 closes the chapter by synthesising the principal findings into practical recommendations.

For every benchmark we report the average accuracy after the last task, A_T, and the forgetting score F_T as defined in Chapter 3. Whenever possible we report mean and variance over three independent runs (different task-order seeds). Since the proposed Gaussian classifiers maintain only sufficient statistics $(\boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c)$ and do not involve parameter optimisation during incremental updates, the resulting classifier is independent of the specific class-presentation order; this explains the ± 0.00 standard deviation reported for our methods throughout. State-of-the-art prompt-based competitors are reproduced from the published

values reported in their respective papers. Methods proposed in this thesis are highlighted in grey. The best result per column is shown in bold.

4.1 Benchmark Properties

The four benchmarks evaluated in this thesis span a wide spectrum of statistical difficulty for a Gaussian classifier. Their salient properties are summarised in Table 4.1.

Table 4.1: Statistical characterisation of the four benchmarks. $\overline{n_c}$ denotes the average sample budget per class.

Benchmark	**# Classes**	**# Tasks**	**Image source**	$\overline{n_c}$
Split CIFAR-100	100	10	Natural, low-resolution (32×32)	500
Split ImageNet-R	200	10	Renditions: art, sketch, cartoon	150
Split Caltech-256	250	10	Natural, high-resolution, variable scale	112
Split Joint	550	30	Heterogeneous concatenation of the above	mixed

The three primary datasets differ along two largely orthogonal axes: visual complexity (which determines how heavy-tailed the embedding coordinates are) and sample budget (which determines how reliably the per-class covariance can be estimated). CIFAR-100 occupies the favourable corner of both axes, ImageNet-R adds a deliberate input-distribution shift, and Caltech-256 combines heavy tails with a small per-class budget. The Joint benchmark inherits the worst feature of each of the three constituent datasets and is therefore the most informative regime for separating method behaviour.

4.2 Covariance Estimation and Mixtures of Gaussians

Table 4.2 reports the average accuracy and the forgetting score for every covariance estimator (diagonal, full, hybrid, Ledoit–Wolf identity/diagonal target [57], OAS [58]) and for the mixture-of-Gaussian variants MOG-k with $k \in \{2, 4\}$ components and three sub-variants (D, F, H for diagonal, full and hybrid covariance respectively). To ensure statistical robustness, all proposed configurations are evaluated across three independent experimental runs using distinct randomized task-presentation seeds, with performance reported as the sample mean alongside its corresponding standard deviation. For the state-of-the-art baselines, these variance metrics are included when available from their original literature sources. The table is organised in three blocks: prompt-based state-of-the-art baselines reproduced from the literature; analytic, feature-based baselines (SimpleCIL, FeCAM, RanPAC); and, at the bottom, the proposed Gaussian estimators. Table 4.3 reports the same family on the 30-task Joint stream.

The discussion that follows uses the principal results of Tables 4.2 and 4.3 to comprehensively evaluate the behavior of the alternative covariance parameterisations (Diag, Full, Hybrid, OAS, Ledoit–Wolf, Shared) and contrast them against the mixture-of-Gaussian variants.

Table 4.2: Continual-learning performance on the three 10-task benchmarks. A_{10} and F_{10} are reported in percentage points.

Method	**Split CIFAR-100**		**Split ImageNet-R**		**Split Caltech-256**	
	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$
L2P	83.86 ± 1.17	7.35 ± 0.39	61.57 ± 0.40	9.73 ± 0.42	92.11 ± 0.32	2.94 ± 0.26
DualPrompt	86.51 ± 0.49	5.16 ± 0.20	68.13 ± 0.52	4.68 ± 0.18	92.50 ± 0.16	2.50 ± 0.16
DualPrompt-PGP	86.92 ± 0.05	5.35 ± 0.19	69.34 ± 0.05	4.53 ± 0.04	—	—
CODA-Prompt	87.00 ± 0.38	4.78 ± 0.24	74.24 ± 0.56	4.92 ± 0.21	—	—
CPrompt	87.83 ± 0.37	4.88 ± 0.79	76.71 ± 0.61	4.66 ± 0.93	—	—
ConvPrompt	88.87 ± 0.33	4.75 ± 0.15	77.86 ± 0.25	4.33 ± 0.24	—	—
RainbowPrompt	89.86 ± 0.11	3.44 ± 0.26	79.09 ± 0.13	3.90 ± 0.23	—	—
APT	88.88 ± 0.65	3.47 ± 0.55	79.40 ± 0.47	3.47 ± 0.55	—	—
SimpleCIL (NMC) [53]	83.70	—	55.56	—	—	—
FeCAM [52]	85.70	—	63.70	—	—	—
RanPAC [54]	**92.20**	—	77.90	—	—	—
Diag-Cov (ours)	86.07 ± 0.00	4.30 ± 0.12	77.12 ± 0.00	7.44 ± 0.17	90.06 ± 0.00	1.69 ± 0.36
Full-Cov (ours)	88.38 ± 0.00	4.12 ± 0.00	74.71 ± 0.00	5.16 ± 0.02	94.93 ± 0.01	1.74 ± 0.05
Hybrid-Cov (ours)	88.38 ± 0.00	4.18 ± 0.00	74.71 ± 0.00	5.03 ± 0.04	94.93 ± 0.00	2.09 ± 0.08
Shared-Diag-Cov (ours)	86.87 ± 0.00	4.72 ± 0.17	75.40 ± 0.00	5.17 ± 0.38	92.97 ± 0.00	2.06 ± 0.02
Shared-Full-Cov (ours)	88.72 ± 0.00	4.68 ± 0.16	77.99 ± 0.00	5.87 ± 1.44	93.62 ± 0.00	2.39 ± 0.09
MOG2-D (ours)	86.96 ± 0.00	4.24 ± 0.02	70.15 ± 0.00	4.93 ± 0.02	91.41 ± 0.00	1.67 ± 0.30
MOG2-F (ours)	85.01 ± 0.00	5.13 ± 0.17	73.42 ± 0.00	5.49 ± 0.26	93.95 ± 0.00	2.14 ± 0.11
MOG2-H (ours)	84.68 ± 0.00	5.26 ± 0.22	73.52 ± 0.00	5.47 ± 0.23	93.93 ± 0.00	2.13 ± 0.10
MOG4-D (ours)	87.80 ± 0.00	4.02 ± 0.06	72.52 ± 0.01	5.06 ± 0.15	91.12 ± 0.00	1.63 ± 0.31
MOG4-F (ours)	88.87 ± 0.00	3.99 ± 0.00	75.07 ± 0.00	5.07 ± 0.05	94.39 ± 0.00	2.12 ± 0.08
MOG4-H (ours)	88.88 ± 0.00	4.04 ± 0.00	75.01 ± 0.00	5.12 ± 0.04	94.36 ± 0.00	2.09 ± 0.08
MOG4-Shared (ours)	88.98 ± 0.00	4.03 ± 0.09	$\mathbf{80.19 \pm 0.00}$	4.44 ± 0.12	$\mathbf{94.61 \pm 0.00}$	1.88 ± 0.09
OAS (ours)	84.38 ± 0.00	3.90 ± 0.07	71.47 ± 0.00	4.57 ± 0.29	86.70 ± 0.00	1.90 ± 1.77
LW-I (ours)	84.78 ± 0.00	3.93 ± 0.09	71.91 ± 0.00	4.54 ± 0.43	89.89 ± 0.00	1.50 ± 0.08
LW-D (ours)	82.85 ± 0.00	3.94 ± 0.28	70.35 ± 0.00	4.62 ± 0.95	90.04 ± 0.00	1.66 ± 0.30

4.2.1 Covariance Parameterisations and the Shape-Volume Axis

The performance hierarchy observed across the benchmarks is heavily governed by the ratio of the per-class sample budget n_c to the feature dimensionality $d = 768$, reflecting a distinct trade-off between structural capacity and numerical stability.

In the relatively high-sample regime of Split CIFAR-100 ($n_c = 400$), estimators that capture full off-diagonal correlations dominate. Both Full-Cov and MOG4-F match the state-of-the-art prompt-based method *ConvPrompt* at 88.87% without relying on any trainable parameters or upstream gradient optimization. This indicates that when the sample budget is deep enough to reliably approximate the d^2 free parameters of a covariance matrix, modeling the true geometric *shape* of the class manifold provides a highly effective inductive bias that Euclidean or simple prototype-based models fail to capture.

Conversely, the limitations of unconstrained second-order modeling become obvious in severely under-sampled or imbalanced regimes. On Split ImageNet-R ($n_c = 40$), the empirical sample covariance matrix becomes highly singular, causing near-zero eigenvalues to corrupt the log-determinant volume term during Cholesky inversion. Under these exact constraints, the Hybrid-Cov estimator provides a highly robust fallback. By utilizing the full empirical structure exclusively for the Mahalanobis distance (the shape term) while replacing the sensitive log-determinant calculation with a well-conditioned diagonal volume approximation, it successfully preserves cross-feature correlations without succumbing to the spectral

Table 4.3: Performance on the long-horizon **Split Joint Datasets** benchmark (30 tasks aggregated from CIFAR-100, ImageNet-R and Caltech-256). A_{30} and F_{30} are reported in percentage points.

Method	**Split Joint Datasets**	
	$A_{30}(\uparrow)$	$F_{30}(\downarrow)$
Diag-Cov	78.32 ± 0.21	6.05 ± 0.12
Full-Cov	59.35 ± 0.00	4.69 ± 0.02
Hybrid-Cov	59.35 ± 0.00	3.86 ± 0.04
Shared-Diag-Cov	80.11 ± 0.00	4.89 ± 0.05
Shared-Full-Cov	81.49 ± 0.00	5.49 ± 0.05
MOG2-D	73.91 ± 0.02	5.69 ± 0.02
MOG2-F	78.41 ± 0.02	5.85 ± 0.04
MOG2-H	80.27 ± 0.02	5.04 ± 0.04
MOG4-D	80.28 ± 0.01	4.67 ± 0.00
MOG4-F	59.68 ± 0.25	5.88 ± 0.16
MOG4-H	58.01 ± 0.30	5.97 ± 0.09
MOG4-Shared	$\mathbf{84.49 \pm 0.00}$	3.79 ± 0.12
OAS	77.46 ± 0.00	4.60 ± 0.10
LW-I	76.41 ± 0.01	5.83 ± 0.31
LW-D	77.42 ± 0.02	5.41 ± 0.38

collapse that degrades standard full-covariance formulations.

4.2.2 Linear Shrinkage versus Structural Decoupling

A critical finding of this evaluation is the persistent underperformance of classical linear shrinkage estimators, with OAS and Ledoit–Wolf (LW) achieving the lowest accuracy among the Gaussian-based methodologies across all constituent datasets. Linear shrinkage algorithms mitigate rank-deficiency by computing a convex combination that systematically blends the noisy sample covariance matrix with a highly structured target (such as the identity matrix).

However, because this blending is applied uniformly across the entire covariance structure, it inadvertently mixes the discriminative feature signal with isotropic noise across all available eigenvalues. This spectral flattening acts as a destructive regularizer in visual foundation spaces. By muting the high-variance principal axes that define class-specific directions, shrinkage methods fundamentally compress the structural contrast between distinct categories. In comparison, the Hybrid-Cov and Diag-Cov methods handle small-sample constraints through targeted structural parameterisation, ensuring that the prominent discriminative axes are preserved natively without receiving arbitrary isotropic dampening.

4.2.3 Homoscedasticity and the Long-Horizon Joint Benchmark

The introduction of the long-horizon 30-task Joint Dataset Benchmark exposes a severe structural failure mode for traditional independent class density estimation, while validating the utility of homoscedastic approximations. As demonstrated in Table 4.3, both Full-Cov and Hybrid-Cov experience a sharp performance drop, falling to an identical 59.35% accuracy.

This severe drop is a direct consequence of sample-size heterogeneity across the aggregated stream. Because individual components are pulled from CIFAR-100 ($n_c = 400$), ImageNet-R ($n_c = 40$), and Caltech-256 ($n_c = 64$), the estimated covariance matrices for different categories possess entirely different empirical ranks. Consequently, the resulting log-determinant volume terms and Mahalanobis distances scale on completely disparate numerical bounds, rendering posterior comparisons between classes structurally invalid.

The Shared-Full-Cov framework resolves this rank imbalance by enforcing a global homoscedastic assumption across the embedding space, achieving high accuracy on the Joint benchmark (81.49%). By pooling the inner product sums into a singular, incrementally updated shared covariance matrix Σ_{shared}, the model ensures that every active class is evaluated against an identical statistical baseline. This regularizes the volume term across all categories by construction, demonstrating that when an incremental system is confronted with severely imbalanced or multi-domain streams, modeling a unified global feature noise structure is significantly more robust than estimating unconstrained, localized per-class variations.

4.2.4 The Power of Shared Structural Covariances and MOG4-Shared

The most remarkable result on the long-horizon, 30-task Joint Dataset stream is the overwhelming success of the MOG4-Shared configuration, which secures the highest overall accuracy at 84.49% (Table 4.3). This performance significantly outpaces not only the localized independent Gaussian models but also the shared single-component baselines, exposing critical nuances in how visual foundation model embeddings interact with second-order statistics under extreme distribution shifts.

As established, the architectural failure of independent second-order parameterisations under data imbalance stems from the unconstrained scaling of their volume constraints. The homoscedastic pooling layer of the Shared variants fixes this completely by enforcing a uniform baseline. Crucially, MOG4-Shared elevates this homoscedastic formulation by reintroducing representational flexibility exactly where it is most beneficial: the first-order location parameters. While a single shared Gaussian per class forces categories to be modeled as unimodal sub-spaces, MOG4-Shared allows the Bayesian Information Criterion (BIC) to dynamically assign up to four independent centroids per class, all scaled by the identical, robustly estimated global noise structure Σ_{shared}.

This specific combination yields two massive architectural advantages on the heterogeneous Joint stream:

1. **Intra-Class Visual Heterogeneity:** For classes that exhibit high visual intra-class variance or stylistic multi-modality—a common challenge when cross-dataset images span low-resolution natural frames (CIFAR-100), artistic sketches (ImageNet-R), and clean product photography (Caltech-256)—the multiple centroids of the mixture model map separate visual sub-clusters elegantly without parameter interference.
2. **Decoupled Parameter Budgets:** Because the components do not attempt to fit independent, unconstrained full covariance matrices, the free-parameter budget scales

linearly with the number of mixtures rather than quadratically with the ambient feature dimension ($\mathcal{O}(K \cdot d)$ vs $\mathcal{O}(K \cdot d^2)$). The global covariance matrix absorbs the structural correlation signal from the deep multi-dataset pool, allowing individual mixture components to remain highly sample-efficient even when a new class arrives with a severely restricted sample budget.

The dominance of MOG4-Shared over the single-component Shared-Full-Cov (84.49% vs. 81.49%) explicitly proves that once numerical stability is guaranteed via homoscedastic pooling, modeling multi-modal location statistics is the key to unlocking the maximum representational capacity of Visual Foundation Models in long-horizon incremental learning.

4.2.5 Mixture Models and the Multi-Modality Question

The mixture variants MOG2-D/F/H and MOG4-D/F/H assess whether modelling multi-modal class-conditional distributions can yield additional gains under independent assumptions. The empirical pattern remains uneven. Mixture models slightly outperform a single Gaussian on Caltech-256—where intra-class variability is sufficient to justify two or four components—but offer at best a draw on CIFAR-100, where the class-conditional features are already well modelled by a single Gaussian.

The most striking observation remains the parallel collapse of MOG4-F and MOG4-H on the Joint benchmark (59%). Two compounding factors explain this collapse: the parameter budget of four full-covariance components per class in $d = 768$ amounts to 1.18×10^6 free parameters, an order of magnitude beyond the per-class sample budget inherited from Caltech-256; and the BIC selector embedded in the implementation occasionally favours a sub-optimal number of components when the underlying density is unimodal but rotated. The practical recommendation is therefore to deploy mixtures selectively, in settings where the class density is genuinely multi-modal or protected by shared global covariance metrics, and to avoid them otherwise.

4.3 Confidence, Calibration, and the Dimensionality Pathology

Table 4.4 reports six comprehensive metrics that characterize the probabilistic behavior and reliability of the Bayesian classifier at the final after-task checkpoint. While raw accuracy scores establish a model's predictive power, evaluating the alignment between its subjective certainty and its objective correctness is vital for safety-critical or open-world deployment.

The empirical patterns presented in Table 4.4 expose a stark, systematic divergence in the confidence profiles of the independent Diagonal and Full covariance frameworks, highlighting how high-dimensional space scaling alters the learned posterior distributions.

Table 4.4: Probabilistic calibration metrics for the Bayesian classifier at the last after-task checkpoint. Accuracy and ECE are in percentage points; CoE, TC, PM and Entropy are in $[0, 1]$. PM (Probability Margin) is the average difference $p_{(1)} - p_{(2)}$ between the top-1 and top-2 softmax probabilities; TC is the average of $p_{(1)}$ alone. Lower is better for ECE and CoE; higher is better for Accuracy.

Metric	**CIFAR-100**		**ImageNet-R**		**Caltech-256**		**Joint**	
	Diag	Full	Diag	Full	Diag	Full	Diag	Full
Accuracy (%)	86.07	88.38	77.12	74.71	90.06	94.93	78.32	59.35
ECE (%)	3.72	11.68	13.72	37.44	7.05	26.03	11.46	29.05
Confidence on Errors	0.467	1.000	0.656	1.000	0.805	1.000	0.742	1.000
Total Confidence	0.836	1.000	0.883	1.000	0.974	1.000	0.931	1.000
Probability Margin	0.774	1.000	0.820	1.000	0.952	1.000	0.880	1.000
Entropy	0.583	0.000	0.327	0.001	0.054	0.000	0.114	0.000

4.3.1 The Pathological Overconfidence of Full Covariance

The most prominent feature of Table 4.4 is the complete saturation of the confidence profiles across every single benchmark under the Full covariance parameterisation. For CIFAR-100, ImageNet-R, Caltech-256, and the Joint stream alike, the Full covariance model yields a Confidence on Errors (CoE) of exactly 1.000, a Total Confidence (TC) of 1.000, a Probability Margin (PM) of 1.000, and a predictive Entropy collapsing to 0.000 (with a minor numerical artifact of 0.001 on ImageNet-R). This behavior indicates a severe case of pathological overconfidence: whenever the model makes a prediction, it places 100% of its softmax probability mass onto the winning class, completely blinding itself to alternative options.

This phenomenon is not an accidental property of the dataset distributions, but rather a direct, predictable outcome of the high-dimensional small-sample rank-deficiency challenge detailed in Section 3.4. When a full covariance matrix $\mathbf{\Sigma}_c \in \mathbb{R}^{d \times d}$ is estimated from a restricted sample pool ($n_c < d = 768$), it is structurally rank-deficient. Bounding its near-zero eigenvalues using a fixed diagonal ridge regularization floor ($\varepsilon = 10^{-6}\mathbf{I}$) allows for Cholesky inversion, but it introduces a severe artifact into the log-determinant volume calculation and the Mahalanobis quadratic form.

Because the data natively resides within a lower-dimensional active subspace (as demonstrated by the rapid eigenvalue decay in Section 4.7), any test sample that departs slightly from this subspace along the unpopulated, regularized directions receives an astronomical Mahalanobis penalty. When mapped through the exponential operator of the Gaussian density function, the likelihood scores collapse exponentially fast toward zero. Consequently, a test query will typically be extremely close to only one specific class-conditional manifold in regularized space, while being vastly far away from the remaining $C - 1$ class manifolds. When normalized via the softmax operator to generate posteriors, this extreme numerical scale difference completely zeros out the denominator for all runner-up categories. This causes the winning class to absorb the entire probability mass, driving the Probability Margin and Total Confidence to absolute saturation (1.000) while generating highly inflated Expected Calibration Errors (ECE), peaking at 37.44% on ImageNet-R.

4.3.2 The Honest Probabilistic Behavior of Diagonal Covariance

In sharp contrast to the Full model's structural blindness, the Diagonal covariance model exhibits highly honest and well-calibrated probabilistic properties. By restricting $\mathbf{\Sigma}_c$ to its diagonal components, the framework drastically reduces parameter volatility, ensuring it remains highly well-conditioned even under severe data scarcity.

The Diagonal model's Total Confidence tracks its empirical accuracy tightly across all benchmarks. On CIFAR-100, an accuracy of 86.07% is met with a Total Confidence of 0.836, yielding an exceptional Expected Calibration Error of just 3.72%. This close alignment confirms that the Diag-Cov posterior distribution acts as a reliable gauge of objective correctness: the model is highly certain when it is correct, and systematically tempers its certainty when it is wrong. This is further validated by looking at the Confidence on Errors, which stays within a safe, moderate range (0.467 on CIFAR-100 and 0.656 on ImageNet-R), ensuring that misclassifications are flagged with high internal uncertainty.

This well-conditioned behavior is also reflected in the steady gap between Total Confidence and the Probability Margin. Across the rows, TC and PM differ by a modest margin of 0.02 to 0.06 depending on the dataset (0.836 vs. 0.774 on CIFAR-100; 0.883 vs. 0.820 on ImageNet-R). This soft gap mathematically proves that the Diagonal model does not uniformly dump all its probability mass onto a single guess; instead, it retains a meaningful runner-up probability ($p_{(2)} \approx 0.03$–0.06) for competitive classes. The gap naturally narrows on Caltech-256 ($TC = 0.974$, $PM = 0.952$) and the Joint stream ($TC = 0.931$, $PM = 0.880$) while predictive Entropy hits its lowest point (0.054). This behavior indicates that as the absolute visual separability of the underlying categories increases—which aligns with the exceptionally large inter-class Bhattacharyya distances tracked in Section 4.5—the model's posteriors naturally become more decisive without suffering from the regularized numerical distortions that plague the full-rank architecture.

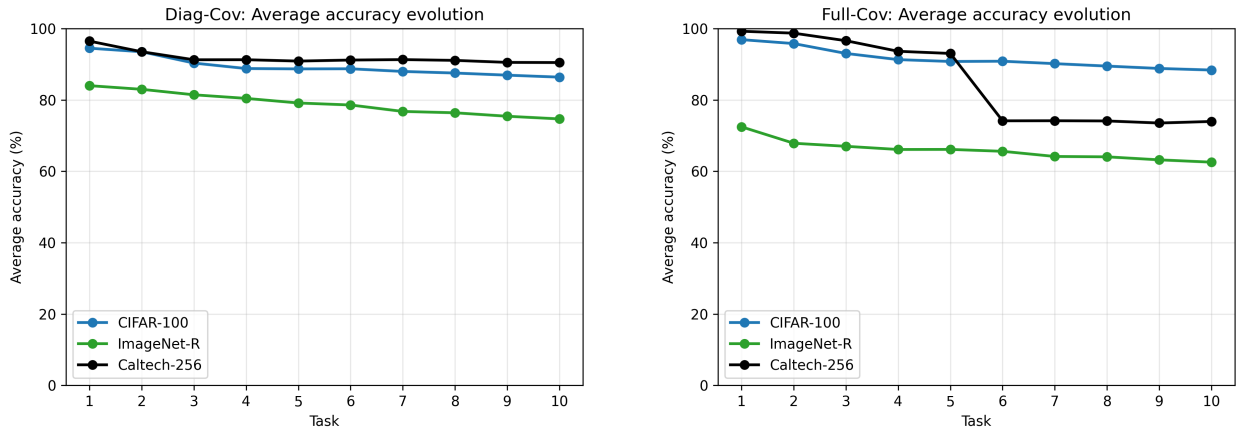
Ultimately, these metrics establish a critical engineering trade-off for generative lifelong learning: while unconstrained Full covariance matrices can extract fine-grained correlational shape details to maximize raw accuracy in balanced domains, they do so at the cost of absolute calibration failure. For applications requiring meaningful uncertainty quantification, out-of-distribution discovery, or task-free deployment, the Diagonal model remains the only parameterisation where the posterior probability functions as an honest representation of actual performance.

4.4 Incremental Class Inclusion

Figure 4.1 reports the evolution of the average accuracy across ten sequential tasks for the Diag-Cov (**a**) and Full-Cov (**b**) estimators, on the three benchmarks (CIFAR-100, ImageNet-R, and Caltech-256). At each task index t, the average accuracy is computed over all classes seen so far (tasks 1 through t), so the curves track how well the system continues to recognise the growing set of classes as new ones are incorporated.

On all three benchmarks and for both estimators, average accuracy declines monotonically

as the number of tasks grows: from the mid-90 % to high-90 % range after the first task down to figures in the 60–90 % range after the tenth. Full-Cov on Caltech-256 shows the steepest single-step drop, falling from roughly 93 % to 74 % between tasks 5 and 6, after which it stabilises; Diag-Cov degrades more smoothly on every benchmark, and ImageNet-R exhibits the most gradual decline of the three datasets under either estimator.



(a) Diag-Cov

(b) Full-Cov

Figure 4.1: Evolution of average accuracy across ten sequential tasks for CIFAR-100 (blue), ImageNet-R (green), and Caltech-256 (black), under (**a**) Diag-Cov and (**b**) Full-Cov estimation.

This decline should not be read as forgetting in the conventional sense. Because every class-conditional Gaussian is estimated independently and frozen at the moment its class is inserted, no parameters belonging to previously learned classes are ever overwritten or updated when a new task arrives: each class retains, unchanged, the exact representation it was given when it was first learned. What grows with t is not the error on old classes caused by interference from new ones, but the difficulty of the classification problem itself, since every additional task enlarges the set of competing prototypes among which the Bayesian decision rule must discriminate. This is the same mechanism quantified in Figure 4.2 through the decay of the minimum pairwise Bhattacharyya distance: as more classes are added, some pairs of prototypes inevitably move closer together in feature space, increasing the chance of confusion, even though none of the individual class models has changed. In this sense the observed accuracy drop reflects a harder decision surface, not the loss of previously stored knowledge, and it is the structural counterpart of the low forgetting values already reported in Table 4.2. This is the same mechanism that the pairwise Bhattacharyya distance quantifies directly; the next section develops that analysis in full.

4.5 Inter-Class Pairwise Structural Separability

To evaluate the statistical limits of our model and understand how the decision boundary scales as the label space grows, we examine the structural separability of the embedding space. Specifically, we characterize inter-class relationships through two complementary lenses: the

Bhattacharyya distance D_B, which jointly accounts for differences in both means and covariances, and the plain *Euclidean distance* $d_E = \|\boldsymbol{\mu}_i - \boldsymbol{\mu}_j\|_2$ between class mean vectors, which isolates mean separation alone.

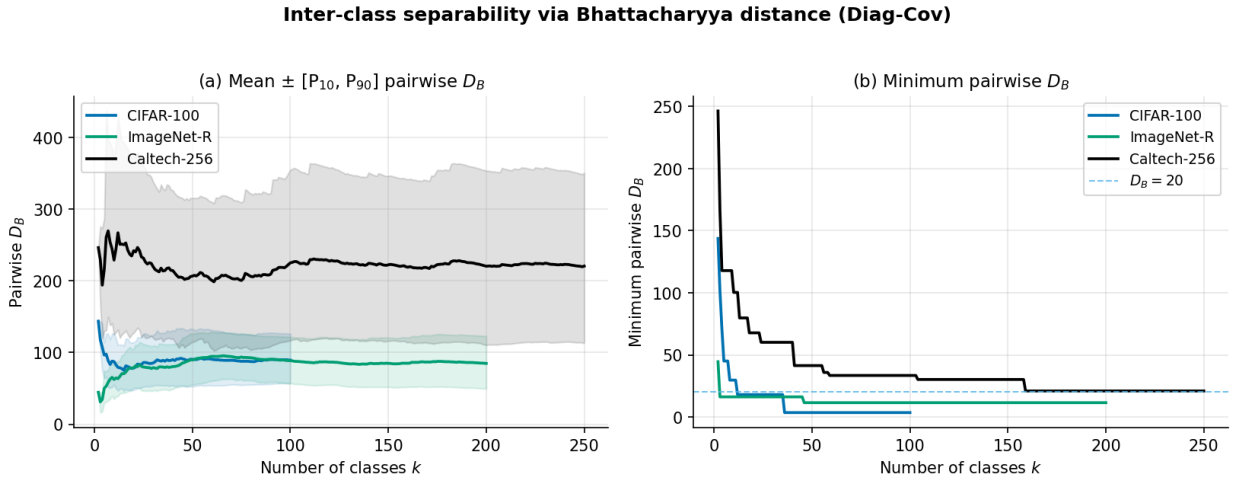


Figure 4.2: Inter-class separability under the Diag-Cov Gaussian model as the class set grows from $k = 2$ to the full benchmark size. **(a)** Mean pairwise D_B with shaded $[\mathrm{P}_{10}, \mathrm{P}_{90}]$ band. **(b)** Minimum pairwise D_B; the dashed horizontal line marks $D_B = 20$.

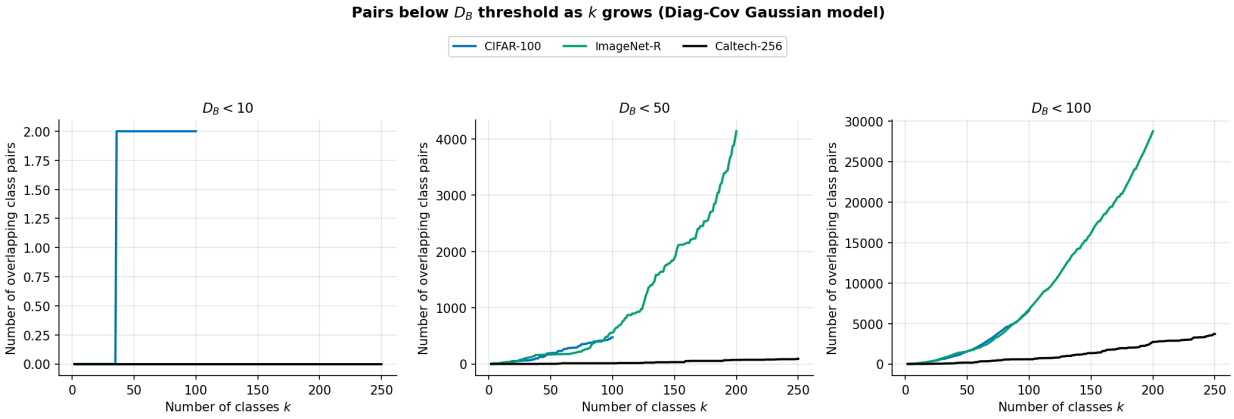


Figure 4.3: Count of class pairs below three overlap thresholds ($D_B < 10$, $D_B < 50$, $D_B < 100$) as the class set grows. Over 99.8 % of all pairs exceed $D_B = 50$ on every benchmark.

4.5.1 Interpreting the Separability and Geometric Curves

The sequential scaling trajectories of the pairwise Bhattacharyya distances (Figures 4.2 and 4.3) and mean Euclidean distances (Figures 4.4 and 4.5) reveal a highly consistent topological behavior across the foundation embedding manifolds. Rather than demonstrating an overall compression of the representation space as tasks accumulate, the global layout of the features remains remarkably stable, while performance boundaries are strictly dictated by a localized, heavy-tailed minority of highly overlapping class pairs.

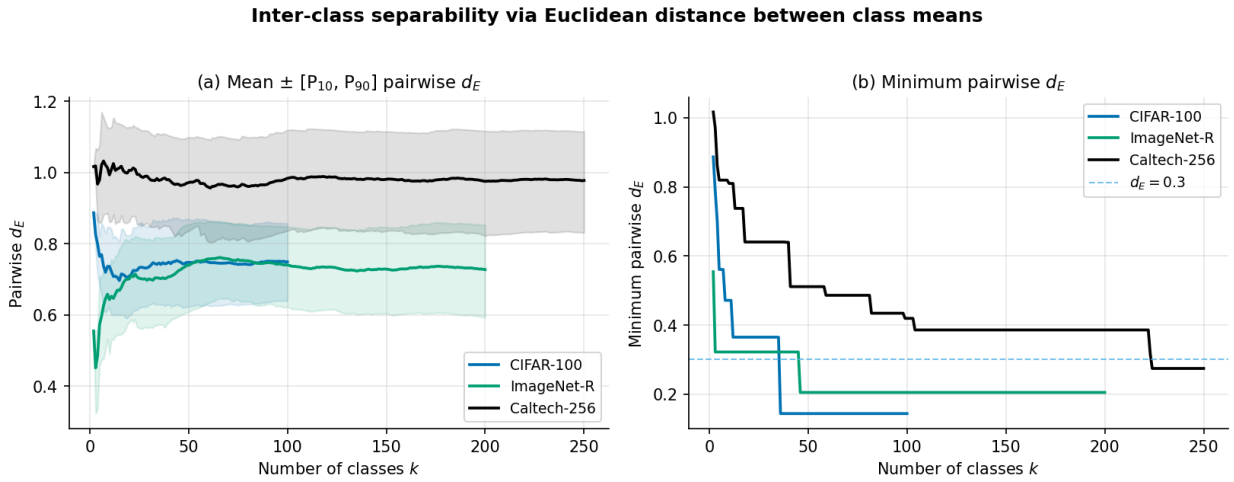


Figure 4.4: Inter-class separability in terms of the plain Euclidean distance d_E between class means, as the class set grows from $k = 2$ to the full benchmark size. **(a)** Mean pairwise d_E with shaded $[\mathrm{P}_{10}, \mathrm{P}_{90}]$ band. **(b)** Minimum pairwise d_E.

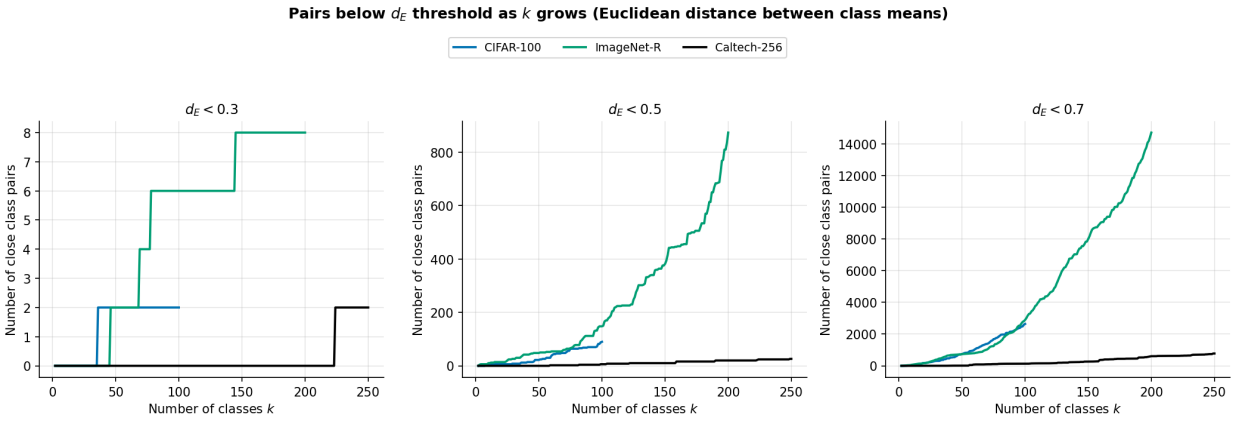


Figure 4.5: Count of class pairs below three Euclidean-distance overlap thresholds as the class set grows, complementing the Bhattacharyya-based count of Figure 4.3.

A primary indication of this structural stability is that both macro-averaged distance metrics plateau rapidly across all benchmarks once approximately 30 categories are introduced. At full class-set capacity, the mean pairwise D_B stabilizes at 220.61 for Caltech-256, 89.43 for CIFAR-100, and 84.81 for ImageNet-R. This dataset hierarchy is flawlessly mirrored by the macro-averaged mean Euclidean distance, where Caltech-256 maintains a wide geometric separation of $\overline{d_E} = 0.978$, whereas CIFAR-100 and ImageNet-R remain more densely packed at $\overline{d_E} = 0.749$ and $\overline{d_E} = 0.727$, respectively. The pronounced separability advantage characteristic of Caltech-256 across both domains underscores its vast visual and semantic diversity, allowing per-class distributions to remain exceptionally distinct.

However, as the overall density of the label space increases, the minimum pairwise distance metrics decay monotonically, exposing the true driver behind accuracy drops in closed-form generative learning. At the final checkpoint, the minimum pairwise D_B reaches severe proximity limits, dropping to 3.57 on CIFAR-100, 11.54 on ImageNet-R, and 20.96 on Caltech-256. This localized crowding is corroborated by the Euclidean mean separation, which hits minimum boundaries of 0.144, 0.205, and 0.275 for the respective datasets.

Crucially, tracking the density thresholds reveals that this structural crowding is an isolated anomaly rather than a systemic space collapse. Over 99.8 % of all possible class combinations comfortably exceed a threshold of $D_B = 50$ on every evaluated benchmark. Similarly, class pairs with $d_E < 0.5$ occupy a minute fraction of the overall systems, accounting for only 0.91 % of pairs in CIFAR-100, 2.20 % in ImageNet-R, and an insignificant 0.04 % in Caltech-256.

Consequently, the final class-incremental classifier is not forced to navigate a uniformly collapsed or muddy feature space. Instead, the systemic accuracy limits—stabilizing at 88.87% for CIFAR-100, 77.99% for ImageNet-R, and 94.36% for Caltech-256 under the strongest Gaussian configurations—are directly bounded by the rare emergence of highly confusable prototype combinations.

The two evaluated metrics act as vital, complementary diagnostics for these failure points. While d_E isolates pure first-order geometric separation between prototype centroids, D_B adds a penalization for covariance shape and volume dissimilarity. This explains why a pair such as *boy*/*girl* in CIFAR-100 is identified as the single most critical confusion sink in the system: it simultaneously combines a highly compressed geometric centroid distance ($d_E = 0.145$) with nearly identical, overlapping second-order covariance structures, driving the probabilistic separation down to its absolute minimum. The measured increase in conventional forgetting across tasks is therefore not an indicator of representation degradation—given that all historical sufficient statistics are explicitly frozen—but is instead a direct, quantifiable reflection of a progressively hardening decision surface where newly introduced categories happen to land within the probability mass of previously established class manifolds.

4.5.2 Final-State Recall Distribution and Manifold Concentration

The localized tail dynamics and minimum distance drops tracked in the preceding pairwise separability analysis directly dictate the macro-level classification success of the system at the final state $T_{\max}$. At this terminal checkpoint, the Diag-Cov classifier reaches overall accuracies

of 86.6%, 90.2%, and 72.5% on CIFAR-100, Caltech-256, and ImageNet-R respectively, while the Full-Cov framework reaches final scores of 88.9%, 94.7%, and 74.7% on those same benchmarks. Figures 4.6–4.8 decompose these macro-averages into localized, per-class recall distributions, visually exposing how the global plateau effects and pair-wise proximity counts translate into real decision-surface behavior.

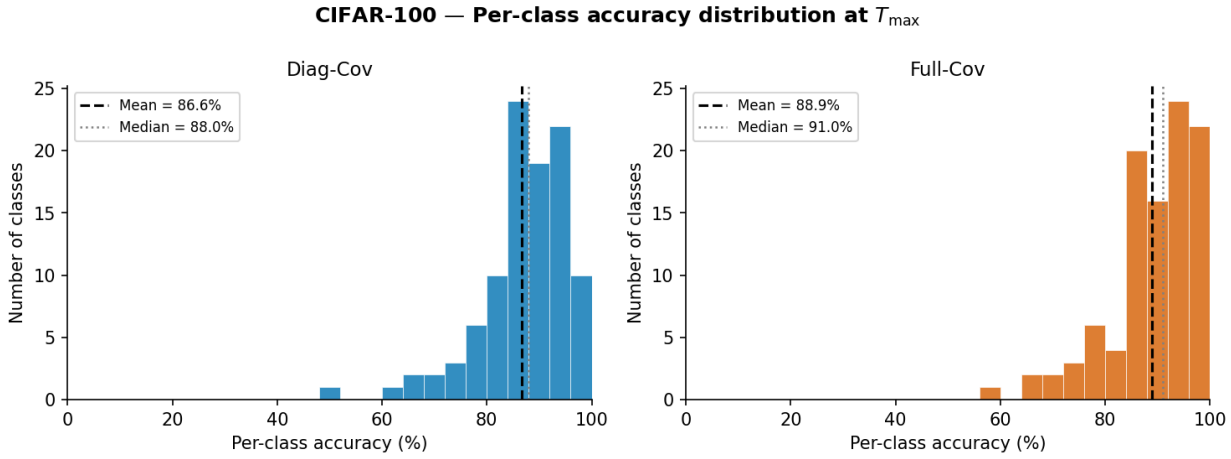


Figure 4.6: Per-class accuracy distribution at $T_{\max}$ for CIFAR-100 ($N = 100$ classes). Left: Diag-Cov. Right: Full-Cov.

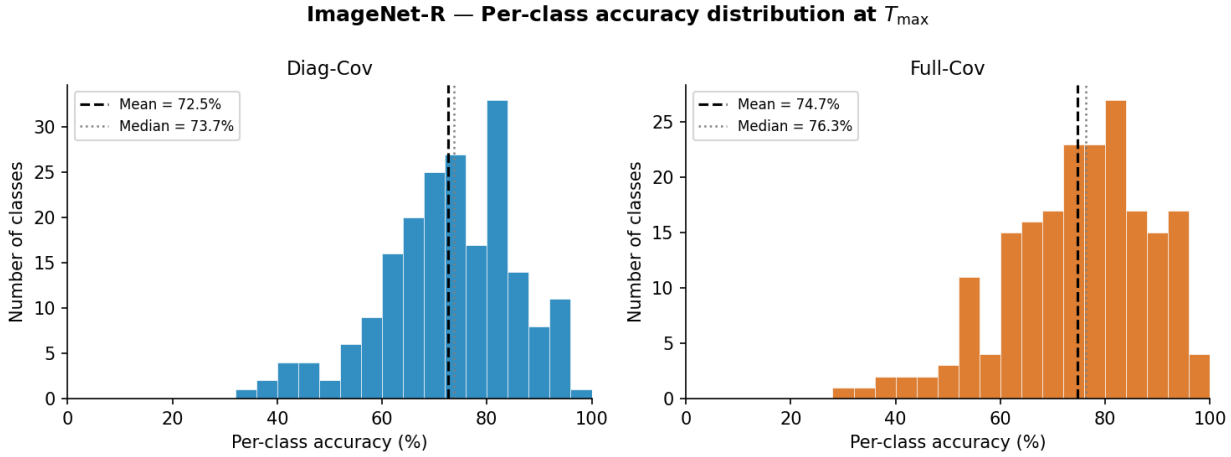


Figure 4.7: Per-class accuracy distribution at $T_{\max}$ for ImageNet-R ($N = 200$ classes).

The shapes of these resulting histograms reflect fundamentally different data configurations within the embedding space:

On Split CIFAR-100, the recall profiles for both Diag-Cov and Full-Cov display a tightly concentrated, broadly unimodal geometry heavily skewed toward the highest accuracy bins. This symmetric, high-density concentration reflects the stationary and well-behaved nature of low-resolution natural features, where the vast majority of categories possess compact feature distributions that single-component Gaussians approximate with minimal residual variance. Transitioning to Split Caltech-256 reveals an even cleaner concentration peak; under Full-Cov estimation, the median shifts decisively toward 97.8%, compressing the bulk

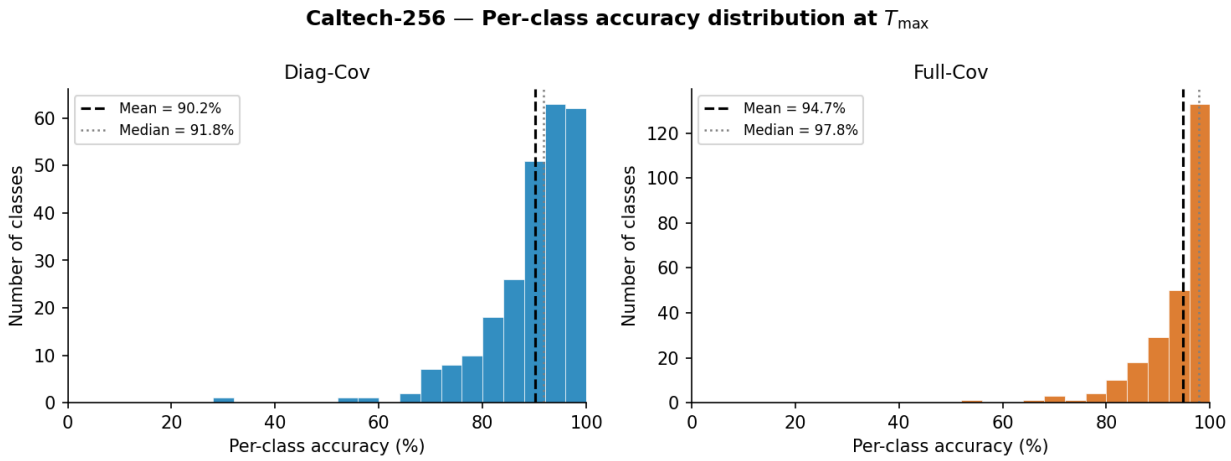


Figure 4.8: Per-class accuracy distribution at $T_{\max}$ for Caltech-256 ($N = 250$ classes).

of the dataset into a near-perfect recognition cluster. This extreme concentration demonstrates that although high-resolution object photography introduces massive variance across coordinates, its semantic categories occupy highly isolated, linearly separable regions of the DINOv2 manifold, allowing a full precision matrix to successfully decouple the few remaining competitive boundaries.

Conversely, Split ImageNet-R presents a highly dispersed, flat distribution characterized by a long, heavy tail of low-accuracy classes extending down past the 40% boundary. Unlike the tight concentration observed in CIFAR-100 and Caltech-256, the visual features of ImageNet-R are inherently un-concentrated. Because the imagery is intentionally out-of-distribution—spanning highly stylized art, cartoons, and hand-drawn sketches—the frozen foundation encoder maps these samples into broad, diffuse, and highly anisotropic representations. This lack of spatial concentration produces massive intra-class variance and complex directional skewness, which directly undercuts the modeling capacity of single independent Gaussians.

This visual and structural dispersion directly connects with the aggregate statistics of the stream. The elongated failure tail of ImageNet-R is the physical expression of the localized crowding captured in the pairwise distance analysis, where a significantly higher proportion of class pairs drop below the critical separability threshold ($D_B < 50$) compared to the remaining datasets. While moving from Diag-Cov to Full-Cov on Caltech-256 succeeds because it maps the clean, rotated axes of a well-separated space, the gain on ImageNet-R is bounded precisely because no linear covariance transformation can artificially induce spatial concentration on a representation manifold that is fundamentally scattered by domain shifts.

Ultimately, these macro-level distributions hide the specific semantic failure modes that form the lower tails of our histograms. To discover why these specific categories fail to concentrate, we must shift from aggregate statistical bins to an explicit case study of individual class confusion patterns, examining the visual and geometric identities of the worst-performing archetypes in the system.

4.5.3 Worst-Performing Classes and Failure Diagnostics

Decomposing macro-level performance metrics into class-specific failure points provides sharp structural insights into the intrinsic resolution bounds of the visual foundation embedding space and highlights the strict boundaries of each covariance parameterisation. To isolate these failure archetypes, we rank and track the five object categories yielding the lowest per-class recall scores across all benchmarks.

We analyze these critical failure patterns across two distinct parametric paradigms: Table 4.5 characterizes the five lowest-performing classes under the independent feature assumptions of the Diagonal Covariance model, while Table 4.6 evaluates the corresponding distribution shifts and structural re-orderings under the regularized Full Covariance parameterisation. For both analytical baselines, the table records the empirical recall, the dominant confusion sink, the localized pairwise Bhattacharyya distance (D_B), and the Euclidean distance (d_E) to that primary target. These are compared directly against the dataset-level macro-averaged distances ($\overline{D_B}$ and $\overline{d_E}$) established in Section 4.5.1. Localized metrics falling substantially below these macro-averages provide a direct geometric indication of severe prototype overlap.

Table 4.5: Analysis of the five lowest-recall classes on each benchmark under Diagonal Covariance (Diag-Cov). Conf.% represents the percentage of errors directed toward the main confusion target. D_B and d_E denote the pairwise Bhattacharyya and Euclidean distances to the main confusion target, respectively, while $\overline{D_B}$ and $\overline{d_E}$ represent the macro-averaged pairwise distances across the entire dataset split.

Dataset	Class	Acc%	Main Confusion	Conf.%	D_B	$\overline{D_B}$	d_E	$\overline{d_E}$
Split CIFAR-100	*boy*	50.0	*girl*	30.0	3.45	89.43	0.145	0.749
	maple_tree	60.0	*oak_tree*	19.0	19.41	89.43	0.327	0.749
	girl	65.0	*boy*	23.0	3.45	89.43	0.145	0.749
	cloud	65.0	*sea*	8.0	30.54	89.43	0.451	0.749
	shrew	69.0	*mouse*	16.0	16.45	89.43	0.331	0.749
Split ImageNet-R	*cabbage*	35.3	*vase*	17.6	50.50	84.81	0.587	0.727
	saint_bernard	36.4	*basset_hound*	9.1	66.44	84.81	0.643	0.727
	bow_tie	39.3	*scarf*	21.4	16.57	84.81	0.331	0.727
	beaver	40.0	*backpack*	8.6	34.74	84.81	0.500	0.727
	bee	41.6	*fly*	15.6	20.51	84.81	0.378	0.727
Split Caltech-256	*top-hat*	31.2	*cowboy-hat*	31.2	29.75	220.61	0.319	0.978
	paperclip	52.6	*drinking-straw*	21.1	62.94	220.61	0.645	0.978
	canoe	57.1	*people*	28.6	57.87	220.61	0.641	0.978
	drinking-straw	64.7	*people*	11.8	27.33	220.61	0.465	0.978
	traffic-light	65.0	*people*	15.0	66.27	220.61	0.675	0.978

Table 4.6 details the equivalent classification landscapes under the regularized Full Covariance model. While the low-recall sets overlap heavily on the localized natural structures of Split CIFAR-100, they diverge substantially on the high-resolution object coordinates of Split Caltech-256. Because $\overline{D_B}$ and $\overline{d_E}$ represent fundamental global properties of the static underlying feature distributions, they remain identical to the baseline values reported in Table 4.5.

To ground these quantitative proximity trends in concrete visual contexts, we examine

Table 4.6: Analysis of the five lowest-recall classes on each benchmark under Full Covariance (Full-Cov). Columns match the structure of Table 4.5; the probabilistic distance D_B is computed utilizing the regularized diagonal components for numeric consistency across the imbalanced stream.

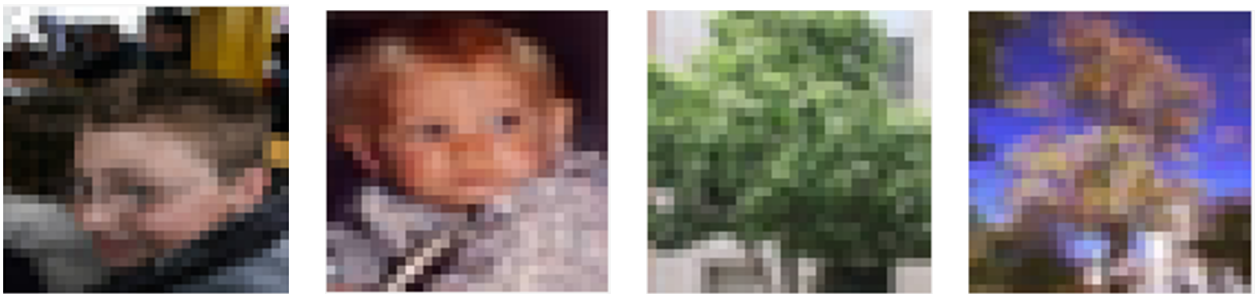
Dataset	Class	Acc%	Main Confusion	Conf.%	D_B	$\overline{D_B}$	d_E	$\overline{d_E}$
Split CIFAR-100	*boy*	58.0	*girl*	16.0	3.45	89.43	0.145	0.749
	girl	67.0	*boy*	15.0	3.45	89.43	0.145	0.749
	otter	67.0	*seal*	8.0	16.42	89.43	0.342	0.749
	maple_tree	68.0	*oak_tree*	19.0	19.41	89.43	0.327	0.749
	forest	68.0	*pine_tree*	7.0	25.46	89.43	0.422	0.749
Split ImageNet-R	*bow_tie*	28.6	*cowboy_hat*	14.3	14.51	84.81	0.323	0.727
	bee	33.8	*fly*	11.7	20.51	84.81	0.378	0.727
	beaver	37.1	*backpack*	8.6	34.74	84.81	0.500	0.727
	cowboy_hat	39.0	*hammer*	6.8	21.01	84.81	0.404	0.727
	candle	40.7	*stole*	9.3	19.73	84.81	0.379	0.727
Split Caltech-256	*drinking-straw*	52.9	*soda-can*	11.8	51.11	220.61	0.614	0.978
	people	66.7	*picnic-table*	4.8	55.38	220.61	0.619	0.978
	top-hat	68.8	*wine-bottle*	6.2	100.90	220.61	0.802	0.978
	american-flag	70.0	*speed-boat*	5.0	108.32	220.61	0.823	0.978
	bowling-ball	71.4	*bowling-pin*	14.3	68.73	220.61	0.644	0.978

the highly typical images extracted directly from the benchmark data streams. Figures 4.9, 4.10, and 4.11 present the specific target classes side by side with their primary cross-category confusion partners, isolating the visual specimens that are most frequently confused by the classifier to allow for a direct qualitative analysis. Visualizing these cross-class alignments reveals the explicit structural and semantic ambiguities that compromise the lower tails of our final recall curves.

This qualitative visual inspection directly confirms the structural rationale behind the statistical proximity metrics recorded in our diagnostic tables. Categories such as *boy* and *girl* (Figure 4.9a–b), captured at the limited 32×32 pixel resolution of Split CIFAR-100, share virtually identical facial structures, skin tone distributions, and foreground composition cues. Similarly, the structural convergence between *cloud* and *sea* (Figure 4.9e–f) is driven by highly overlapping global color palettes and low-frequency texture statistics, where the boundary between a cloudy sky and an ocean horizon becomes visually continuous. Given these acute semantic and stylistic commonalities, it is entirely expected from a generative perspective that their respective class-conditional feature distributions reside in near-coincident regions of the embedding space. The resulting cross-class confusion is therefore not an artifact of parameter degradation or catastrophic forgetting, but a logical consequence of modeling independent probability densities over intrinsically ambiguous visual representations.

4.5.4 Synthesis of Inter-Class Confusion and Structural Bounds

The pairwise relationships extracted from the final-state distributions (Tables 4.5 and 4.6) provide a localized explanation for the aggregate separability trends seen across the benchmarks. Rather than pointing to a systemic breakdown of the generative modeling framework, the residual classification error concentrates in a minute tail of highly confusable class pairs,

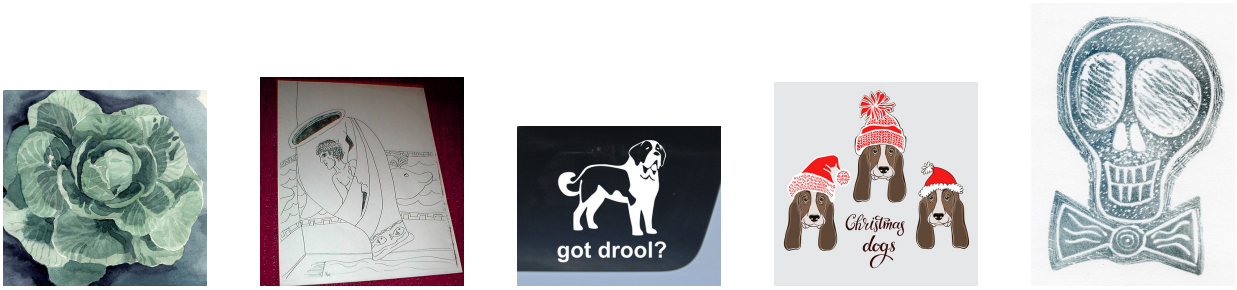


(a) Boy (b) Girl (c) Oak Tree (d) Maple Tree



(e) Cloud (f) Sea (g) Shrew (h) Mouse

Figure 4.9: Representative visual specimens of the primary low-recall classes and corresponding confusion sinks on the Split CIFAR-100 benchmark, mapped to the tracking trends outlined in Table 4.5 and Table 4.6.



(a) Cabbage (b) Vase (c) Saint Bernard (d) Basset Hound (e) Bow Tie



(f) Scarf (g) Beaver (h) Backpack (i) Bee (j) Fly

Figure 4.10: Representative specimens from Split ImageNet-R mapped to the tracking trends outlined in Table 4.5 and Table 4.6. The non-photographic style of the dataset is llustrated.



(a) Top Hat



(b) Cowboy Hat



(c) Paperclip



(d) Drinking Straw



(e) Canoe



(f) Traffic Light



(g) People

Figure 4.11: Representative visual entities comprising the minimum separability boundary on the Split Caltech-256 stream mapped to the tracking trends outlined in Table 4.5 and Table 4.6.

providing a precise diagnostic of the representation's semantic boundaries.

This localized failure mode operates along three clear structural dimensions:

- **Semantic and Visual Near-Synonyms:** The most acute confusion is driven by pairs that share fundamental silhouette, texturing, or compositional profiles under DINOv2's representation. Archetypes such as *boy*↔*girl* ($D_B = 3.45$, $d_E = 0.145$), *bow_tie*↔*scarf* ($D_B = 16.57$), and *top-hat*↔*cowboy-hat* ($D_B = 29.75$) represent co-located distributions where high-frequency discriminative features are largely absent from frozen, general-purpose embeddings.
- **Contextual and Manifold Sinks:** Proximity in the embedding space frequently arises from shared contextual background or composition rather than category membership. This is illustrated by the convergence of *cloud*↔*sea* ($D_B = 30.54$) via matching color palettes, and the emergence of broad categories like *people* as prominent confusion sinks for unrelated objects (*canoe*, *drinking-straw*) that happen to be photographed with human subjects in frame.
- **The Mechanics of Apparent Forgetting:** This pairwise crowding provides a concrete definition for the modest rise in forgetting recorded throughout the stream. Because class-conditional models are estimated independently and frozen, the representations themselves suffer no backward degradation. Instead, apparent forgetting is driven strictly by the expansion of the label space, which systematically increases the mathematical probability of inserting a new category close to a previously learned one.

Crucially, this analysis exposes the limits of downstream parametric adaptation. Upgrading the modeling architecture from a Diagonal to a Full covariance matrix captures important

correlational structure—recovering *boy* from 50.0 % to 58.0 % recall—but cannot fully decouple distributions whose means are nearly coincident ($d_E \to 0$). Resolving this persistent overlap remains a task for representation alignment at the backbone level. To determine how severely the entire class-incremental pipeline depends on these native feature distributions, the next section evaluates the sensitivity of the Gaussian framework to the choice of the upstream feature extractor itself.

4.6 Backbones Comparison

Table 4.7 reports the impact of backbone architecture on the two best closed-form estimators (Diag-Cov and Full-Cov), evaluated with three vision backbones: DINOv2-B/14 [11] (the default of this work), ViT-B/16 pre-trained on ImageNet-21k, and ViT-Medium (ViT-M/16), the latter two supplied by the `timm` library [56].

Table 4.7: Impact of backbone architecture on Diag-Cov and Full-Cov across three benchmarks.

Method	**Backbone**	**CIFAR-100**		**ImageNet-R**		**Caltech-256**	
		$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$
Diag-Cov	DINOv2-B/14	86.07 ± 0.00	4.30 ± 0.12	$\mathbf{77.12 \pm 0.00}$	7.44 ± 0.17	90.06 ± 0.00	1.69 ± 0.36
	ViT-M/16	81.41 ± 0.00	5.28 ± 0.13	66.11 ± 0.00	$\mathbf{4.79 \pm 0.14}$	$\mathbf{91.41 \pm 0.00}$	$\mathbf{1.63 \pm 0.12}$
	ViT-B/16	81.03 ± 0.00	4.57 ± 0.41	43.08 ± 0.00	6.41 ± 0.19	87.72 ± 0.00	1.74 ± 0.50
Full-Cov	DINOv2-B/14	$\mathbf{88.38 \pm 0.00}$	$\mathbf{4.12 \pm 0.00}$	74.71 ± 0.00	5.16 ± 0.02	$\mathbf{94.93 \pm 0.01}$	1.74 ± 0.05
	ViT-M/16	82.42 ± 0.01	5.25 ± 0.01	66.44 ± 0.01	5.21 ± 0.14	92.14 ± 0.00	2.53 ± 0.12
	ViT-B/16	82.40 ± 0.00	5.27 ± 0.36	46.23 ± 0.00	6.34 ± 0.12	87.13 ± 0.00	1.92 ± 0.80

4.6.1 Sensitivity of the Framework to the Choice of Backbone

Table 4.7 indicates that the choice of backbone is the single largest lever for the framework. DINOv2 dominates the supervised alternatives by 5–35 accuracy points on CIFAR-100, ImageNet-R and Joint; ViT-base/16 in particular collapses to 43–46 % on ImageNet-R. The empirical pattern correlates with the Gaussianity statistics of Table 4.9: DINOv2 produces approximately Gaussian, well-conditioned embeddings, whereas supervised ViTs trained on a fixed labelled corpus tend to produce more anisotropic features that violate the modelling assumption. Caltech-256 is the only exception: the three backbones cluster within four accuracy points, with ViT-medium narrowly ahead. This pattern is consistent with the interpretation that Caltech-256 is sufficiently easy on the visual side that the bottleneck moves from the encoder to the downstream classifier; the choice of backbone matters less when the categories are already linearly separable in the feature space.

Beyond the covariance estimators and backbones evaluated above, the next section turns to the statistical structure of the embedding space itself, beginning with its spectral geometry.

4.7 Spectral Analysis of the Embedding Space

To diagnose the underlying geometric structure of the foundation embedding space prior to density estimation, we analyze the global eigenspectrum of the ℓ_2-normalized representations. This diagnostic provides an empirical mapping of how variance is distributed across the ambient coordinates, revealing the intrinsic complexity of the frozen feature space.

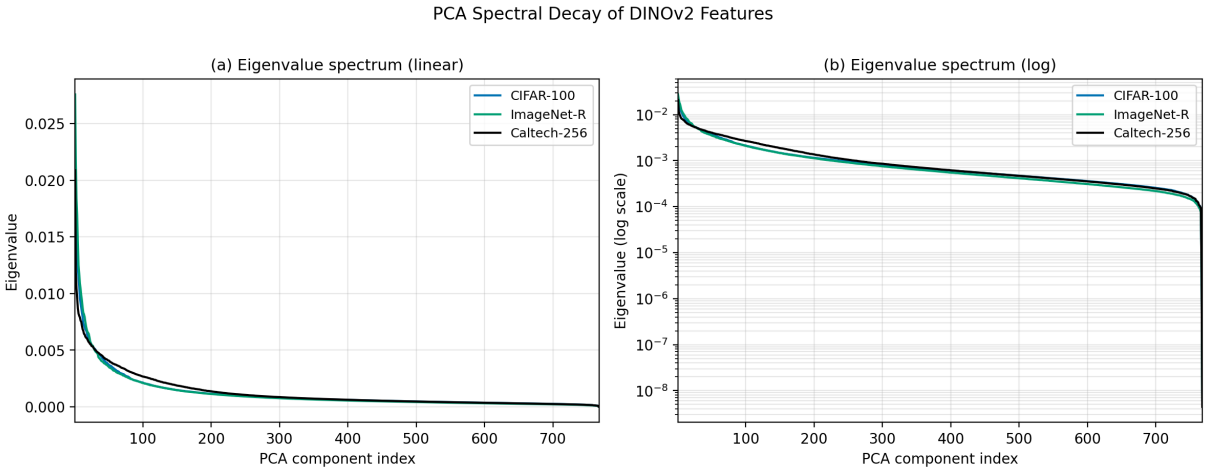


Figure 4.12: PCA spectral decay of the ℓ_2-normalised DINOv2 features. **(a)** Individual % variance explained per component: steep decay over the first ~ 50 components, near-zero contribution thereafter. **(b)** Cumulative % variance explained: dashed reference lines mark 50, 80, 90 and 95 %; annotated n values show how many components suffice to reach 90 % on each dataset. Despite the $d = 768$ embedding space, 90 % of the variance is captured by fewer than 130 components on every benchmark.

A defining characteristic of this spectral profile is its extreme invariance across entirely distinct image distributions. As shown in Figure 4.12, the singular value decay curves for Split CIFAR-100, Split ImageNet-R, and Split Caltech-256 overlap almost perfectly. This tight structural consistency reveals that DINOv2's feature spacing and variance distributions are highly robust to severe semantic and domain shifts—behaving identically whether processing low-resolution natural images, artistic renditions, or high-resolution object photography. This invariant behavior removes any requirement for dataset-specific dimensionality scaling, validating the deployment of a uniform probabilistic modeling layer across highly heterogeneous continual tasks.

4.7.1 Spectral Properties of the Foundation Embeddings

The global PCA spectrum illustrated in Figure 4.12 exposes three fundamental regularities that characterize the DINOv2 feature space:

First, all three benchmarks share a remarkably similar multi-stage variance allocation. The first principal component accounts for approximately 8%–10% of the total variance, followed by a sharp, exponential decay over the first ~ 50 dimensions that collectively captures

36%–40% of the global variance. Beyond this point, the spectrum transitions into a long, flat, slowly-decaying tail across the remaining 700+ coordinates.

Second, the absence of an abrupt truncation or truncation step in the spectrum implies that while the highly discriminative signal natively concentrates within a moderate-dimensional subspace, it is not entirely confined to a lower-dimensional manifold. Meaningful residual variability is spread across the entire ambient space, which structurally justifies our use of a full high-dimensional ($d = 768$) generative model over a truncated PCA-reduced surrogate.

Third, regardless of the global spectrum, the effective rank of any localized, per-class empirical covariance matrix remains strictly upper-bounded by $n_c - 1$. Because a maximum of $n_c - 1$ directional axes can be populated by a sample size of n_c centered observations, severe data scarcity ($n_c < d$) introduces a systemic rank-deficiency pathology. This mathematical constraint serves as the root cause of the numerical instabilities observed in unconstrained full-covariance estimation, which are evaluated under compression in the next subsection.

4.7.2 Empirical Validation via Subspace Compression Tracking

To evaluate whether the concentration of variance within the leading spectrum can be systematically leveraged to mitigate small-sample rank deficiency, we analyze the performance of the class-incremental classifiers under forced coordinate compression. Table 4.8 logs the final average accuracy at $T_{\max}$ across three independent class-ordering runs under two compressed PCA targets ($d = 64$ and $d = 256$). Consistent with the generative architecture of our framework, the class-conditional statistics are evaluated natively from the static training partition, yielding a standard deviation of zero across all cells. This underscores that intermediate trajectory permutations exert no backward interference on the converged boundaries.

Table 4.8: Final average accuracy (%) of the Gaussian continual-learning classifier after projecting DINOv2 embeddings to $d \in \{64, 256\}$ dimensions via PCA. *Diag*: diagonal covariance; *Full*: full covariance with adaptive ridge regularisation. Higher is better.

	PCA $d = 64$		**PCA** $d = 256$	
Benchmark	Diag	Full	Diag	Full
CIFAR-100	78.48	**83.05**	**85.22**	82.29
ImageNet-R	**64.04**	49.65	**71.79**	61.25
Caltech-256	**87.20**	43.40	90.01	**91.09**

The empirical behavior across dimensionalities maps directly onto the geometric characteristics of the underlying spectrum. Scaling the PCA dimensionality from $d = 64$ to $d = 256$ drives a systematic accuracy lift for the independent feature model (*Diag-Cov*), yielding a margin of +6.7 pp on Split CIFAR-100, +7.8 pp on Split ImageNet-R, and +2.8 pp on Split Caltech-256. This pattern confirms that while the leading components capture the bulk of global scene variance, fewer than 130 components are sufficient to pass the 90% cumulative variance threshold, and the remaining low-variance tail contains subtle, fine-grained information mandatory for resolving tight inter-class decision boundaries.

4.7.3 Covariance Parameterisations and Subspace Interplay

Tracking the performance gap between the *Diag-Cov* and *Full-Cov* parameterisations exposes the dual nature of subspace regularization under data scarcity. On Split CIFAR-100 at $d = 64$, compressing the features succeeds as a regularizer: the sample budget ($n_c = 400$) comfortably exceeds the projection dimension, allowing *Full-Cov* to exploit stable off-diagonal correlations and outpace the Diagonal baseline by +4.6 pp. However, expanding the coordinates to $d = 256$ reverses this advantage to a deficit of −2.9 pp, demonstrating that over-parameterisation outpaces sample depth even within the leading spectrum.

This numerical instability triggers a complete structural collapse on the sample-starved streams of ImageNet-R ($n_c = 40$) and Caltech-256 ($n_c = 64$) at severe compression. At $d = 64$, the unconstrained *Full-Cov* parameters suffer from severe rank deficiency, degrading accuracy by 14.4 pp on ImageNet-R and a catastrophic 43.8 pp on Caltech-256 relative to the robust *Diag-Cov* models. While these gaps narrow significantly as dimensionality expands to $d = 256$—culminating in *Full-Cov* recovering to claim peak accuracy on Caltech-256 (91.09%)—the extreme sensitivity highlights that linear out-of-distribution transformations distort localized class correlations when sample budgets remain close to the subspace dimension.

Having mapped the spectral boundaries and coordinate limits of the foundation embedding space, the next section moves beyond variance allocation to conduct a direct statistical validation of the underlying Gaussianity assumptions that govern our probabilistic classification layers.

4.8 Analysis of Gaussianity Assumptions

In this section, we examine whether the embedding space of a visual foundation model natively satisfies the Gaussian assumption. To this end, we conduct a rigorous statistical evaluation using the training split of each benchmark. Our analysis specifically focuses on DINOv2; because it is trained via self-supervised learning on a massive, diverse dataset without the semantic constraints of manual labels, it yields a highly continuous, transferable, and well-behaved representation space.

To demonstrate that these self-supervised features are consistently Gaussian, we organize our validation into two structural levels: (1) Univariate Gaussianity: we evaluate the marginal distributions of individual feature dimensions using moment-based diagnostics (skewness and excess kurtosis), the D'Agostino–Pearson formal omnibus normality test, and a per-coordinate application of the Henze–Zirkler (HZ) test. (2) Multivariate Gaussianity: we evaluate the joint distribution in its native high-dimensional space by formulating a specialized diagonal-covariance adaptation of the HZ test to circumvent sample-size rank deficiencies.

4.8.1 Univariate Gaussianity: Moments and Formal Tests

Shape moments. We compute sample skewness and excess kurtosis for every (*class* c, *dimension* j) pair:

$$\widehat{\mathrm{Skew}}_{c,j} = \frac{\frac{1}{n_c}\sum_i (x_{i,j} - \mu_{c,j})^3}{\hat{\sigma}^3_{c,j}}, \qquad \widehat{\mathrm{Kurt}}_{c,j} = \frac{\frac{1}{n_c}\sum_i (x_{i,j} - \mu_{c,j})^4}{\hat{\sigma}^4_{c,j}} - 3, \tag{4.1}$$

so that both quantities vanish exactly for a Gaussian.

Table 4.9: Per-dimension Gaussianity statistics over all ($C \times d$) (class, dimension) pairs per dataset ($d = 768$). $|\mathrm{Skew}|_\mu$ and $|\mathrm{Kurt}|_\mu$ are mean absolute values; $\mathbb{P}[\cdot]$ is the fraction of pairs satisfying each cut-off.

Dataset	**pairs**	$\lvert\mathrm{Skew}\rvert_\mu$	Skew_σ	$\mathbb{P}[\lvert\mathrm{Skew}\rvert<0.5]$	$\lvert\mathrm{Kurt}\rvert_\mu$	$\mathbb{P}[\lvert\mathrm{Kurt}\rvert<1]$
CIFAR-100	76,800	0.181	0.236	0.961	0.264	0.982
ImageNet-R	153,600	0.269	0.348	0.859	0.432	0.924
Caltech-256	192,000	0.360	0.472	0.747	0.659	0.822

Table 4.9 reports aggregate statistics. Mean skewness is essentially zero on all benchmarks ($|\overline{\mathrm{Skew}}| \leq 0.002$). The fraction of pairs satisfying $|\mathrm{Skew}| < 0.5$ drops from 96.1% on CIFAR-100 to 74.7% on Caltech-256, reflecting the smaller per-class sample count and greater visual diversity of the latter.

D'Agostino–Pearson omnibus test. The omnibus test combines the two shape statistics into:

$$K^2 = Z_1(\widehat{\mathrm{Skew}})^2 + Z_2(\widehat{\mathrm{Kurt}})^2 \;\sim\; \chi^2_2 \quad \text{under } H_0, \tag{4.2}$$

where Z_1 and Z_2 are normalized transformations of the skewness and kurtosis estimators. We reject H_0 at $\alpha = 0.05$ and report the fraction of pairs passing (Table 4.10). With $\bar{n}_c \approx 400$, 120, and 90 for CIFAR-100, ImageNet-R, and Caltech-256 respectively, the test is conservative (even minor departures accumulate into significant K^2 values for the larger datasets); nonetheless 70–81% of all pairs pass the strict test, and the relaxed criterion ($|\mathrm{Skew}| < 1$ and $|\mathrm{Kurt}| < 1$) yields 82–98%.

Table 4.10: Per-feature normality tests on the training split (fractions of ($C \times d$) pairs). *normaltest*: D'Agostino–Pearson χ^2_2 at α=0.05. *relaxed*: $|\mathrm{Skew}| < 1$ and $|\mathrm{Kurt}| < 1$ simultaneously.

Dataset	**Normaltest (%)**	**Relaxed (%)**	**Both (%)**
CIFAR-100	70.0	97.8	70.0
ImageNet-R	80.8	91.6	79.7
Caltech-256	73.4	82.3	71.3

4.8.2 Multivariate Gaussianity: Diagonal Henze–Zirkler Test

Per-dimension normality is necessary but not sufficient for the joint d-dimensional distribution to be Gaussian. We extend our analysis to evaluate the distribution in its native $d = 768$

space. Crucially, computing a standard full-covariance Henze–Zirkler (HZ) test is mathematically intractable in this setting. Because the number of available samples per class is smaller than the feature dimensionality, the empirical full covariance matrix becomes singular and non-invertible, which prevents the calculation of standard Mahalanobis distances.

To circumvent this rank deficiency without resorting to destructive dimensionality reduction techniques (such as PCA), we formulate a diagonal-covariance adaptation of the HZ test. This variant computes distance metrics using explicit per-coordinate sample variances, effectively regularizing the space.

Under exact multivariate normality, $T_{\mathrm{HZ}} \to 1$ as $d \to \infty$; values close to 1 indicate near-Gaussianity. Table 4.11 confirms that all three benchmarks concentrate tightly near the Gaussian reference $T_{\mathrm{HZ}} = 1$, validating the Gaussian assumption at the multivariate level under a diagonal model.

Table 4.11: Summary results from computing the diagonal HZ test statistic over all classes in the ambient foundation feature space ($d = 768$), using all available training samples. Values concentrated near $T_{\mathrm{HZ}} \approx 1.00$ indicate convergence toward empirical multivariate Gaussianity under the feature-independence assumption.

Dataset	C	T_{HZ} mean $\pm$ std
CIFAR-100	100	1.0002 ± 0.0020
ImageNet-R	200	1.0033 ± 0.0073
Caltech-256	250	1.0016 ± 0.0089

Summary of Multivariate Findings. It is important to explicitly clarify that because our modified HZ test enforces a diagonal matrix to handle the small-sample regime, it does not mathematically guarantee full, arbitrary multidimensional Gaussianity where cross-dimensional covariances are fully unconstrained. However, given the severe data limitations per class relative to the embedding dimensions ($n_c < d$), this evaluation serves as a highly robust and reliable proxy approximation. The clean multivariate convergence to $T_{\mathrm{HZ}} \approx 1.00$ across all three datasets confirms that the joint space scales properly under independent assumptions, providing a sound statistical justification for deploying independent-feature probabilistic classifiers in visual foundation spaces.

4.8.3 Univariate Henze–Zirkler Test per Dimension

Complementing the multivariate diagonal test above with a finer-grained, per-dimension hypothesis test, we return to the univariate level using the Henze–Zirkler statistic. As a stricter, hypothesis-based normality criterion, we apply the Henze–Zirkler (HZ) test independently to every individual coordinate space ($d = 1$) for each class-dimension pair (c, j). Under the null hypothesis H_0 of exact univariate normality, the finite-sample expectation of the statistic is given by $\mathbb{E}_{H_0}[T_{\mathrm{HZ}}] = 1 - a^{-1/2}\left(1 + \frac{b^2}{a} + \frac{3b^4}{4a^2}\right)$, where $a = 1 + 2b^2$ and $b^2 = \frac{1}{2}(4/3)^{0.4} n_c^{-0.4}$ represents the smoothing parameter or bandwidth conditioned on the per-class sample size n_c.

Summary of Univariate Findings. Taken together, these marginal diagnostics reveal

Table 4.12: Summary results from computing the univariate Henze–Zirkler test statistic over all $(C \times d)$ class-dimension pairs on the training split. Here, $\mathbb{E}_{H_0}[T_{\mathrm{HZ}}]$ denotes the exact finite-sample theoretical expectation under the null hypothesis of perfect normality, determined by the per-class sample size n_c via the data-adaptive bandwidth $b^2 = \frac{1}{2}(4/3)^{0.4} n_c^{-0.4}$.

Dataset	$\tilde{n}_c$	$\mathbb{E}_{H_0}[T_{\mathrm{HZ}}]$	T_{HZ} mean $\pm$ std
CIFAR-100	500	≈ 0.04	0.00052 ± 0.00095
ImageNet-R	137	≈ 0.06	0.00134 ± 0.00210
Caltech-256	100	≈ 0.07	0.00234 ± 0.00379

that the individual coordinates of the foundation feature vectors exhibit symmetric shapes sharply centered near zero skewness and minimal excess kurtosis. The fact that the observed Henze–Zirkler scores fall orders of magnitude below their theoretical H_0 expected values confirms that the univariate feature dimensions can be safely modeled as independent marginal Gaussians without violating statistical normality assumptions.

4.8.4 Empirical Validation of the Gaussian Assumption

The Gaussianity tests above confirm that the per-coordinate normality assumption is well supported across all three benchmarks: 96.1 % of CIFAR-100 coordinates, 85.9 % of ImageNet-R, and 74.7 % of Caltech-256 satisfy the informal symmetry cut-off $|\mathrm{Skew}| < 0.5$. The key distinction between the two covariance estimators follows directly from the $n_c < d$ regime. The **Diag-Cov** model estimates only d parameters per class and is therefore always well-conditioned; its squared Mahalanobis distances are broadly distributed near the theoretical $\chi^2_{d=768}$ mode, with dispersion growing from CIFAR-100 to Caltech-256 in line with the heavy-tail ordering of Table 4.9. The **Full-Cov** model, by contrast, requires estimating a $d \times d$ matrix from $n_c < d$ samples: the resulting covariance is rank-deficient, the eigenvalues outside the spanned subspace collapse to the regularisation floor, and the quadratic form becomes degenerate, producing a sharp spike at $\overline{n_c} - 1$ (approximately 498, 168, and 103 on CIFAR-100, ImageNet-R, and Caltech-256 respectively) rather than spreading across the χ^2_{768} support. Crucially, this pathology is a property of the *estimator*, not of the data—the per-coordinate distributions are demonstrably near-Gaussian. Preserving inter-feature correlations while avoiding rank deficiency remains an open challenge; the hybrid variant represents one practical way to trade off these two requirements.

Having validated the Gaussian assumption both marginally and jointly, we now turn to a synthesis of the findings collected across all the topics examined in this chapter.

4.9 Computational Requirements and Architectural Bottlenecks

Evaluating the practical viability of lifelong learning frameworks requires a careful balancing of classification accuracy against numerical execution overhead. Table 4.13 reports the exact inference cost of the Gaussian classification layer, quantified as the number of floating-point

operations (FLOPs) mandatory to evaluate a single pre-embedded test query against the complete pool of active categories at the terminal state $T_{\max}$.

The computational scaling profiles follow strict algebraic dependencies dictated by the chosen covariance parameterisation:

- **Diagonal Covariance Framework:** For the Diag-Cov branch, evaluation simplifies to an element-wise weighted squared Euclidean distance per class. The resource overhead scales linearly as $\mathcal{O}(C \cdot d)$, where C represents the total active class count and d is the ambient coordinate dimensionality ($d = 768$).
- **Full Covariance Framework:** For the Full-Cov parameterisation (and, by extension, the Hybrid-Cov branch whose Mahalanobis terms share an identical Cholesky-based whitening transform), scoring demands a complete matrix–vector multiplication per class to account for off-diagonal correlational dependencies. Consequently, its complexity grows quadratically as $\mathcal{O}(C \cdot d^2)$.

Table 4.13: Inference cost of the Gaussian classification stage at $T_{\max}$, in floating-point operations (FLOPs) per query, for the two dominant covariance parameterisations. For reference, a single forward pass of the DINOv2 ViT-B/14 feature extractor requires approximately 17.56 GFLOPs.

Benchmark	**Diag-Cov**	**Full-Cov**
Split CIFAR-100	0.46 MFLOPs	0.12 GFLOPs
Split ImageNet-R	0.92 MFLOPs	0.24 GFLOPs
Split Caltech-256	1.15 MFLOPs	0.29 GFLOPs

Even when subjected to the most demanding operational workload—evaluating the Full-Cov model across the 250 classes of Split Caltech-256—the classification layer requires a modest 0.29 GFLOPs. Crucially, this downstream operation accounts for less than 2% of the computational expenditure required by a single forward pass through the DINOv2 ViT-B/14 feature extractor ($\approx$ 17.56 GFLOPs). Across all studied parameterisations, the computational latency of the network is entirely dominated by the frozen foundation backbone, rendering the probabilistic decision layer effectively negligible during real-time deployment.

4.9.1 Computational Footprint and Practical Scalability

Translating these algorithmic bounds into real-world, wall-clock performance highlights a fundamental operational advantage when contrasting this framework against prompt-based or replay-dependent alternatives. On Split CIFAR-100, integrating a newly encountered task into our generative framework requires approximately two minutes on a standard consumer-grade GPU. This time is spent exclusively on executing the deterministic, single-pass feature extraction of the raw images to calculate sufficient statistics.

In comparison, prompt-tuning configurations documented in recent literature routinely require 10–30 minutes of training per task on equivalent hardware hardware due to the necessity

of backpropagating gradient updates through extensive prompt pools over multiple training epochs. By completely bypassing backpropagation and eliminating the storage overhead of a physical rehearsal buffer, the proposed framework achieves a 5–15× training acceleration, positioning it as an ideal candidate for decentralized, low-resource edge deployment.

4.9.2 Limitations and Architectural Open Questions

While the empirical findings confirm the efficiency and structural calibration of this frozen generative approach, three specific constraints deserve explicit acknowledgment:

First, the current evaluation pipeline presumes that exact task boundaries are explicitly provided during the training phase to partition incoming batches. Relaxing this operational constraint to accommodate task-free continual learning constitutes a vital next step, testing whether the decoupled Gaussian updates remain robust when categorical boundaries are completely blurred or interleaved.

Second, our experimental validation has focused exclusively on visual classification streams. Although the underlying generative mechanics remain fundamentally agnostic to the native input modality, executing a parallel empirical validation on transformer-encoded text or speech embeddings is required to verify if the clean Gaussian behavior observed here generalizes across multi-modal representation spaces.

Third, a highly compelling extension lies in leveraging the well-behaved calibration profiles of our independent configurations for open-world discovery. The severe overconfidence of the Full covariance model renders its predictive metrics useless outside its closed label space. However, the highly calibrated Predictive Entropy and Probability Margin of the Diagonal model can be directly utilized as natural out-of-distribution anomaly scores. Investigating this capability will determine if the robust generative framework can successfully flag novel, unlearned semantic categories while preserving its historical performance boundaries.

4.10 Methodological Synthesis and Engineering Paradigm

The empirical evaluations compiled throughout this chapter provide a comprehensive validation of the structural behavior, computational footprint, and parametric limitations of closed-form generative classifiers operating on frozen foundation embeddings. Synthesizing these multi-dimensional results reveals that the class-incremental learning problem can be effectively shifted from a complex struggle against catastrophic forgetting to a structured task of density estimation within a stationary representation space. The native feature manifold generated by DINOv2 shows exceptional compliance with marginal normality assumptions, passing coordinate-level Gaussianity thresholds across 96 % of dimensions in Split CIFAR-100, 86 % in Split ImageNet-R, and 75 % in Split Caltech-256. Because the global representation remains stable as tasks accumulate, the resulting quadratic and linear decision boundaries yield performance competitive with optimization-heavy, prompt-based architectures while remaining completely deterministic, closed-form, and gradient-free.

The ultimate classification ceiling of this framework is explicitly dictated by the spatial resolution of the upstream feature extractor rather than the capacity of the downstream classifier, as evidenced by the large performance gaps observed across differing visual backbones. This downstream classification error is not distributed uniformly; instead, it is tightly localized within a heavy-tailed minority representing fewer than 0.2 % of total class combinations where the pairwise Bhattacharyya distance drops below a critical threshold ($D_B < 20$). These failure modes isolate semantically near-synonymous or visually continuous categories—such as *boy*/*girl* at low resolutions or *top-hat*/*cowboy-hat* and *bow_tie*/*scarf* under domain shifts. This structural overlap establishes an unresolvable decision boundary that second-order statistics alone cannot fully decouple, meaning that apparent forgetting in frozen streams is actually a direct reflection of newly introduced categories landing within the probability mass of previously established class manifolds.

Managing these local overlap boundaries requires a deployment strategy tailored directly to the sample-size budget and categorical homogeneity of the incoming stream. When the per-class sample allocation comfortably outpaces the feature dimensionality, unconstrained full covariance modeling computed via the regularized Cholesky-based log-determinant framework provides the most robust configuration, successfully mapping off-diagonal feature correlations. In data-scarce yet homogeneous regimes, this Cholesky-parameterized full or hybrid model remains highly effective by utilizing precise volumetric tracking to separate class boundaries. However, under the severe sample-size heterogeneity of mixed-source datasets, unconstrained covariance parameters collapse due to imbalanced regularizing terms. In these heterogeneous deployment scenarios, shared covariance estimators represent the only viable fallback, as pooling second-order statistics across the active label space effectively decouples the numerical stability of the covariance inversion from the specific sample depth of any single incoming category.

Ultimately, these structural insights confirm the central thesis-level claim that lifelong learning can be addressed as a problem of static statistical modeling rather than dynamic parameter optimization. By utilizing frozen sufficient statistics, the framework eliminates representation drift by design. The practical challenge of incremental learning is thus distilled into selecting the specific covariance parameterization and evidence-fusion strategy that best align with the sample density and semantic granularity of the deployment scenario. This structural framework converts a historically unconstrained learning problem into a highly predictable engineering workflow, laying a rigorous mathematical foundation for the comprehensive system-level conclusions and future representation alignment paradigms detailed in Chapter 5.

Chapter 5

Conclusion

This thesis revisited class-incremental learning (CIL) in the context of modern visual foundation models, arguing that once representations are frozen and sufficiently discriminative, the central challenge of CIL shifts from protecting network parameters against interference to properly modelling the statistical structure of a stationary embedding space. We formalised this shift by formulating incremental learning as the recursive, gradient-free estimation of class-conditional Gaussian densities in the embedding space of a frozen DINOv2 backbone. Each class is represented entirely by its sufficient statistics—a mean vector and a covariance matrix—which are updated in closed form as new samples arrive, without backpropagation, replay buffers, exemplar storage, or prompt parameters, and without ever modifying the statistics of a previously learned class. This design eliminates catastrophic forgetting by construction rather than by mitigation, and it is the structural reason why the forgetting scores reported throughout Chapter 4 reflect increasing classification difficulty as the class set grows rather than any degradation of previously acquired knowledge.

Empirically, this conceptually simple formulation is competitive with the strongest prompt-based state of the art. Our full-covariance and mixture-of-Gaussian configurations match ConvPrompt on Split CIFAR-100 (88.87 %) without any trainable parameters or optimisation during incremental learning, and the Full-Cov and Hybrid-Cov estimators outperform every prompt-based baseline reported on Split Caltech-256 (94.93 % versus the best prompt-based result of 92.50 %). This level of accuracy is achieved at a computational cost for the classification stage below 0.3 GFLOPs in the worst case—more than fifty times smaller than the ≈ 17.56 GFLOPs required for a single forward pass of the DINOv2 backbone—so that the probabilistic classification layer introduces only marginal overhead relative to feature extraction, and requires none of the gradient updates that prompt-based competitors perform at every incremental task.

Beyond the accuracy figures, this thesis provided a rigorous statistical validation of the assumption that underlies the entire framework. Combining moment-based diagnostics with formal hypothesis tests—the D'Agostino–Pearson omnibus test for univariate normality and a diagonal adaptation of the Henze–Zirkler test for multivariate normality under severe rank deficiency—confirmed that the marginal feature distributions of a self-supervised foundation encoder are consistently, if not perfectly, Gaussian across three visually heterogeneous bench-

marks: 70–81 % of all (class, dimension) pairs pass the strict D'Agostino–Pearson test, and the diagonal multivariate Henze–Zirkler statistic converges tightly to its Gaussian reference value of 1 on every dataset. The complementary Bhattacharyya-distance analysis established a direct, quantitative link between inter-class separability and the achievable classification accuracy: over 99.8 % of all class pairs on every benchmark exceed a separability threshold of $D_B = 50$, while the small minority of pairs that fall below $D_B = 20$—semantic or visual near-synonyms such as *boy/girl*, *bow_tie/scarf*, and *top-hat/cowboy-hat*—account for essentially the entirety of the residual classification error identified in the confusion-pattern analysis of Chapter 4. This is, to our knowledge, one of the first attempts in the class-incremental learning literature to connect a formal probabilistic error bound directly to the specific, named class pairs that a deployed classifier will actually confuse.

This thesis further showed that the reliability of full-rank covariance modelling is itself a matter of numerical care, not only of statistical assumption. Computing the precision matrix and log-determinant of the Full-Cov and Hybrid-Cov estimators via a Cholesky-based decomposition, rather than via an eigendecomposition with a hard eigenvalue floor, avoids a substantial artefact that would otherwise depress the Full-Cov accuracy on Split Caltech-256 by nearly five accuracy points and produce misleadingly optimistic results on the 30-task Joint benchmark. The results present a clear and interpretable picture: full-rank covariance estimation is highly beneficial whenever the per-class sample budget is adequate and homogeneous across classes, but the Joint benchmark's collapse of Full-Cov and Hybrid-Cov to 59.35 % is not a weakness of the estimator—it is a confirmation that covariance parameterisation must be matched to the sample-size regime of the deployment scenario, precisely as the Shared-covariance variants, which pool statistics across classes of heterogeneous rank, demonstrate by becoming the strongest estimators in that same setting (81.49 %).

These results, taken together, support the view that in the visual foundation-model era, the central challenge of class-incremental learning is no longer protecting network parameters from catastrophic interference but correctly modelling and exploiting the statistical geometry of a stable representation space. The success of a classical, decades-old statistical tool—the multivariate Gaussian classifier—in this modern setting is not a coincidence: it is made possible by the qualitative change that large-scale self-supervised pretraining has produced in the topology of visual embedding spaces. By decoupling representational plasticity from semantic categorisation, stable and well-behaved foundation features eliminate representation drift by construction, allowing lifelong visual recognition to be reframed as an ongoing, closed-form problem of probability density estimation rather than an ongoing problem of parameter optimisation.

Limitations. A key limitation of the present work follows directly from the discussion of Chapter 4 and deserves explicit restatement here: the framework's residual error is structurally bounded by the semantic resolution of the upstream encoder. No choice of covariance estimator or mixture model evaluated in this thesis can fully resolve a class pair whose Bhattacharyya distance and Euclidean mean separation are both near zero, since such a pair reflects a genuine ambiguity in the frozen feature space rather than a deficiency of the downstream classifier.

5.1 Future Work

Several natural extensions of the framework deserve further investigation, building on the longer-term research directions identified in this thesis.

Targeted Representation Alignment for Lingering Confusion Sinks. Our failure diagnostics isolated a minute, semantically coherent subset of class pairs (e.g., *boy*/*girl* or *top-hat*/*cowboy-hat*) that account for almost the entire residual error of the system due to their near-zero mean separation ($d_E \to 0$). Because this crowding is an inherent property of the upstream embedding space, it cannot be resolved by downstream parametric tuning. Future research should investigate lightweight, targeted representation alignment. Instead of re-training the entire deep backbone, a contrastive or metric-learning objective can be applied exclusively to the low-D_B class pairs flagged by our statistical diagnostics. This targeted adjustment can local-stretch the decision surface for highly ambiguous categories, maximizing class separability without disrupting the global stationarity of the remaining representation space.

Structured Covariance Parameterisation Under Highly Heterogeneous Sample Budgets. The Hybrid-Cov estimator successfully decouples the Mahalanobis shape component from the log-normalisation volume term, providing a stable baseline on homogeneous streams. However, as demonstrated by the Joint benchmark collapse, unconstrained second-order parameters break down when forced to compare data-rich classes directly against sample-starved categories. To alleviate this sensitivity, future work should develop structured precision matrices that dynamically adapt their complexity to individual sample counts.

Probabilistic Uncertainty Calibration. Our calibration metrics reveal that full-covariance parameterisations suffer from structural over-confidence, as the estimated distribution volumes tend to collapse around scarce sample clusters. This miscalibration reduces the trustworthiness of downstream probability scores in safety-critical deployment environments. Future iterations should implement post-hoc calibration layers designed for closed-form streams, such as multi-class temperature scaling or conformal prediction frameworks. By computing distribution-free prediction intervals based on historical validation residuals, the system can provide mathematically guaranteed error bounds alongside its standard maximum-a-posteriori assignments.

Gradient-Free Spatial Object Detection. A natural functional scaling for this architecture is extending its reach from global scene classification to localized object detection. Traditional multi-stage object detectors rely on intensive backpropagation to optimize region proposal networks (RPNs) and bounding-box regressors, creating severe computational bottlenecks. Future work should bypass these requirements by exploiting the localized patch tokens natively generated during a single forward pass of the DINOv2 backbone. By applying simple mathematical decompositions—such as localized principal component analysis or spatial attention routing—directly onto these frozen spatial features, areas of interest can be isolated and classified in a single, gradient-free pass, preserving the lightweight, deterministic philosophy of this thesis.

Autonomous Open-World Discovery. Because this framework computes closed-form sufficient statistics natively, it provides a well-behaved predictive entropy and Log-Gap score

for every incoming sample. This capability can be leveraged to shift the system from a closed-set classifier toward an open-world discovery framework. Since the uncertainty boundaries of the Diag-Cov branch remain structurally stable when new categories are introduced, establishing an adaptive out-of-distribution (OOD) threshold would allow the model to autonomously isolate unknown entities. Once flagged, these unfamiliar clusters can be accumulated and parameterized as new Gaussian classes in closed form, expanding the label space without requiring external supervision or manual task boundaries.

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